

Supplementary material for *couplr: Optimal Matching with Verifiable Certificates and Sparse Edge Generation*

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1 What this file holds

The article reports summaries of four measurements that are too large to print in full: the solver grid over problem regimes, the edge-generation loop over covariate counts, point clouds, calipers and sizes, the peak memory of every mode and comparator, and the raw runs behind the scaling table. This file carries them in full, and names the script that produced each one so any of them can be re-measured. It also carries the parts of the software design the article points at: the object model, the C++ organisation and the dependency footprint.

```
## run      20260828-170754
## host     Darwin Mac 25.5.0 Darwin Kernel Version 25.5.0: Mon Apr 27 20:41:15 PDT 2026; root:xnu-12
## commit   5a1693c 40 modified paths
## R version 4.5.3 (2026-03-11)
## aarch64-apple-darwin25.3.0
## couplr   1.6.2
## MatchIt  4.7.2
## optmatch 0.10.8
## microbenchmark 1.5.0
```

Section	Script	Files
Solver selection across regimes	scripts/bench_regimes.R	regime-runs.csv, regime-cells.csv, regime-results.csv, regime-verdict.csv
The edge-generation loop	scripts/bench_implicit_grid.R	implicit-grid-runs.csv, implicit-grid-results.csv
Peak memory	scripts/bench_memory.R	memory-results.csv
Representation against solver	scripts/bench_implicit.R	implicit-results.csv
Scaling, every run	scripts/bench_scaling.R	scaling-runs.csv, scaling-results.csv

Every script is resume-safe: it writes its rows as it goes and skips a cell it already holds, so a run that is interrupted continues where it stopped. Moving the CSV aside forces a full re-measurement. `paper/run_bench_suite.sh` runs all of them in order on one machine, which is how the numbers here were produced.

2 Solver selection across problem regimes

`assignment(method = "auto")` chooses a solver from five rules read off one pass over the cost matrix: enumerate an at most 8 by 8 problem, use a cardinality algorithm where the finite costs are constant or binary, use the sparse Jonker-Volgenant variant where more than half the entries are forbidden, use the shortest-augmenting-path solver where there are at least three times as many columns as rows, and otherwise use Jonker-Volgenant. The article's solver figure varies problem size on square dense integer costs, which is the last rule's regime and no other. This grid varies the properties the rules are stated over.

A cell is a cost regime, a forbidden pattern and a shape. Each cell is generated as several independent instances, each instance is timed several times, and every solver in the panel sees the same instance.

Table 1: The factors the solver grid crosses. Shapes and sizes are in the next table.

Factor	Level	Meaning
cost regime	int_uniform	integer, uniform on 1..10,000
cost regime	dbl_uniform	double, uniform on (0, 1)
cost regime	binary	binary, 0 or 1
cost regime	constant	constant
cost regime	tied	integer, five distinct values
cost regime	heavy_tailed	double, lognormal
cost regime	metric_uniform	euclidean over uniform points in 8 dimensions
cost regime	metric_clustered	euclidean over eight clusters in 8 dimensions
forbidden pattern	none	complete: every pair admissible
forbidden pattern	random_60	60 percent of pairs admissible, unstructured
forbidden pattern	random_25	25 percent admissible, unstructured
forbidden pattern	block_4	25 percent admissible in four disconnected components

Table 2: Shapes. The base tier runs the whole panel at a size every solver in it reaches; the large tier repeats the crossing at a size only the scalable solvers reach.

tier	Shape	aspect	Solvers in the panel
base	500x500	1:1	12
base	500x1500	1:3	12
base	500x5000	1:10	12
large	1000x10000	1:10	5
large	1500x1500	1:1	5
large	1500x4500	1:3	5

2.1 What the rules cost

For each cell, the solver "auto" selected is timed as a solver of its own, so its number carries the probing pass the rules cost, and it is compared against the fastest named solver in the same cell.

Table 3: Each dispatch rule, over the cells where it fired. Ratio is the time the dispatched solver took over the time the cell's fastest named solver took, so 1.00x means the rule picked the best available solver and 2.00x means it cost twice the best.

Rule	Cells	Picked the cell's fastest	Median ratio	Worst ratio
sparse	48	2	1.16x	12.44x
very_rectangular	32	0	5.75x	13.38x
no_cost_scale	30	24	1.00x	1.02x
default	16	10	1.00x	10.28x

Table 4: The 25 cells where the dispatched solver was furthest off the cell's fastest.

Regime	Pattern	Shape	Rule	Picked	Best	Dispatched (ms)	Best (ms)	Ratio
tied	complete	500x1500	very_rectangular	sap	network_simplex	273	20.4	13.38x
tied	25%	500x500	sparse	lapmod	auction_scaled	73	5.9	12.44x
tied	complete	500x500	default	jv	network_simplex	57.1	5.6	10.28x
metric_uniform	complete	500x5000	very_rectangular	sap	jv	110	11	10.02x
metric_clustered	complete	500x5000	very_rectangular	sap	jv	107	11.1	9.68x
heavy_tailed	complete	500x5000	very_rectangular	sap	hungarian	100	11.2	9.00x
dbl_uniform	complete	500x5000	very_rectangular	sap	jv	102	11.4	8.94x
int_uniform	complete	500x5000	very_rectangular	sap	hungarian	101	11.9	8.50x
tied	60%	500x500	default	jv	network_simplex	62.5	7.5	8.28x
tied	complete	500x5000	very_rectangular	sap	network_simplex	864	118	7.29x
metric_uniform	complete	500x1500	very_rectangular	sap	jv	26.8	3.7	7.24x
heavy_tailed	complete	500x1500	very_rectangular	sap	jv	26.7	3.7	7.13x
dbl_uniform	complete	500x1500	very_rectangular	sap	hungarian	26.8	3.9	6.82x
int_uniform	complete	500x1500	very_rectangular	sap	hungarian	27.2	4.1	6.57x
metric_clustered	complete	500x1500	very_rectangular	sap	jv	25.4	3.9	6.50x
tied	25%	500x1500	sparse	lapmod	network_simplex	222	35.3	6.29x
dbl_uniform	complete	1000x10000	very_rectangular	sap	jv	475	77.1	6.16x
int_uniform	complete	1000x10000	very_rectangular	sap	jv	474	78.6	6.02x

Regime	Pattern	Shape	Rule	Picked	Best	Dispatched (ms)	Best (ms)	Ratio
metric_clustered	complete	1000x1000	very_rectangular	sap	jv	464	77	6.02x
heavy_tailed	complete	1000x1000	very_rectangular	sap	jv	465	77.5	6.00x
tied	60%	500x1500	very_rectangular	sap	network_simplex	195	35.5	5.49x
dbl_uniform	complete	1500x4500	very_rectangular	sap	jv	320	59	5.43x
metric_clustered	complete	1500x4500	very_rectangular	sap	jv	316	59.2	5.34x
int_uniform	complete	1500x4500	very_rectangular	sap	jv	319	60.6	5.27x
heavy_tailed	complete	1500x4500	very_rectangular	sap	jv	319	62.8	5.08x

2.2 Every cell

Table 5: Every cell of the solver grid: the rule that fired, the solver it picked, the fastest solver in the cell, and the ratio between them.

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
binary	4 blocks	500x1500	no_cost_scale	hk01	hk01	1.00x
binary	4 blocks	500x500	no_cost_scale	hk01	hk01	1.01x
binary	4 blocks	500x5000	no_cost_scale	hk01	hk01	1.00x
binary	complete	500x1500	no_cost_scale	hk01	hk01	1.00x
binary	complete	500x500	no_cost_scale	hk01	hk01	1.01x
binary	complete	500x5000	no_cost_scale	hk01	hk01	1.00x
binary	25%	500x1500	no_cost_scale	hk01	hk01	1.00x
binary	25%	500x500	no_cost_scale	hk01	hk01	1.00x
binary	25%	500x5000	no_cost_scale	hk01	hk01	1.00x
binary	60%	500x1500	no_cost_scale	hk01	hk01	1.01x
binary	60%	500x500	no_cost_scale	hk01	hk01	1.02x
binary	60%	500x5000	no_cost_scale	hk01	hk01	1.00x
constant	4 blocks	500x1500	no_cost_scale	hk01	hk01	1.00x
constant	4 blocks	500x500	no_cost_scale	hk01	hk01	1.00x
constant	4 blocks	500x5000	no_cost_scale	hk01	hk01	1.00x
constant	complete	500x1500	no_cost_scale	hk01	hk01	1.01x
constant	complete	500x500	no_cost_scale	hk01	hk01	1.00x
constant	complete	500x5000	no_cost_scale	hk01	hk01	1.01x
constant	25%	500x1500	no_cost_scale	hk01	hk01	1.00x
constant	25%	500x500	no_cost_scale	hk01	hk01	1.00x
constant	25%	500x5000	no_cost_scale	hk01	hk01	0.99x
constant	60%	500x1500	no_cost_scale	hk01	hk01	1.00x
constant	60%	500x500	no_cost_scale	hk01	hk01	1.00x
constant	60%	500x5000	no_cost_scale	hk01	hk01	1.00x
dbl_uniform	4 blocks	500x1500	sparse	lapmod	hungarian	1.17x
dbl_uniform	4 blocks	500x500	sparse	lapmod	auction_scaled	1.60x
dbl_uniform	4 blocks	500x5000	sparse	lapmod	hungarian	1.14x
dbl_uniform	complete	500x1500	very_rectangular	sap	hungarian	6.82x
dbl_uniform	complete	500x500	default	jv	jv	1.01x
dbl_uniform	complete	500x5000	very_rectangular	sap	jv	8.94x
dbl_uniform	25%	500x1500	sparse	lapmod	jv	1.15x
dbl_uniform	25%	500x500	sparse	lapmod	auction_scaled	2.42x
dbl_uniform	25%	500x5000	sparse	lapmod	hungarian	1.18x
dbl_uniform	60%	500x1500	very_rectangular	sap	jv	2.94x
dbl_uniform	60%	500x500	default	jv	auction_scaled	1.19x
dbl_uniform	60%	500x5000	very_rectangular	sap	jv	3.06x

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
heavy_tailed	4 blocks	500x1500	sparse	lapmod	jv	1.16x
heavy_tailed	4 blocks	500x500	sparse	lapmod	hungarian	1.17x
heavy_tailed	4 blocks	500x5000	sparse	lapmod	jv	1.13x
heavy_tailed	complete	500x1500	very_rectangular	sap	jv	7.13x
heavy_tailed	complete	500x500	default	jv	jv	1.00x
heavy_tailed	complete	500x5000	very_rectangular	sap	hungarian	9.00x
heavy_tailed	25%	500x1500	sparse	lapmod	jv	1.13x
heavy_tailed	25%	500x500	sparse	lapmod	auction_scaled	1.64x
heavy_tailed	25%	500x5000	sparse	lapmod	hungarian	1.18x
heavy_tailed	60%	500x1500	very_rectangular	sap	jv	2.93x
heavy_tailed	60%	500x500	default	jv	jv	1.00x
heavy_tailed	60%	500x5000	very_rectangular	sap	jv	3.08x
int_uniform	4 blocks	500x1500	sparse	lapmod	hungarian	1.16x
int_uniform	4 blocks	500x500	sparse	lapmod	auction_scaled	1.48x
int_uniform	4 blocks	500x5000	sparse	lapmod	hungarian	1.12x
int_uniform	complete	500x1500	very_rectangular	sap	hungarian	6.57x
int_uniform	complete	500x500	default	jv	jv	1.02x
int_uniform	complete	500x5000	very_rectangular	sap	hungarian	8.50x
int_uniform	25%	500x1500	sparse	lapmod	hungarian	1.15x
int_uniform	25%	500x500	sparse	lapmod	auction_scaled	1.73x
int_uniform	25%	500x5000	sparse	lapmod	hungarian	1.19x
int_uniform	60%	500x1500	very_rectangular	sap	hungarian	2.95x
int_uniform	60%	500x500	default	jv	auction_scaled	1.08x
int_uniform	60%	500x5000	very_rectangular	sap	hungarian	3.08x
metric_clustered	4 blocks	500x1500	sparse	lapmod	hungarian	1.16x
metric_clustered	4 blocks	500x500	sparse	lapmod	auction_scaled	2.17x
metric_clustered	4 blocks	500x5000	sparse	lapmod	jv	1.15x
metric_clustered	complete	500x1500	very_rectangular	sap	jv	6.50x
metric_clustered	complete	500x500	default	jv	jv	1.00x
metric_clustered	complete	500x5000	very_rectangular	sap	jv	9.68x
metric_clustered	25%	500x1500	sparse	lapmod	jv	1.12x
metric_clustered	25%	500x500	sparse	lapmod	auction_scaled	3.56x
metric_clustered	25%	500x5000	sparse	lapmod	jv	0.87x
metric_clustered	60%	500x1500	very_rectangular	sap	jv	2.90x
metric_clustered	60%	500x500	default	jv	auction_scaled	1.79x
metric_clustered	60%	500x5000	very_rectangular	sap	jv	3.01x
metric_uniform	4 blocks	500x1500	sparse	lapmod	jv	1.18x
metric_uniform	4 blocks	500x500	sparse	lapmod	auction_scaled	1.96x
metric_uniform	4 blocks	500x5000	sparse	lapmod	hungarian	0.74x
metric_uniform	complete	500x1500	very_rectangular	sap	jv	7.24x
metric_uniform	complete	500x500	default	jv	jv	1.00x
metric_uniform	complete	500x5000	very_rectangular	sap	jv	10.02x
metric_uniform	25%	500x1500	sparse	lapmod	jv	1.14x
metric_uniform	25%	500x500	sparse	lapmod	auction_scaled	3.11x
metric_uniform	25%	500x5000	sparse	lapmod	jv	1.16x
metric_uniform	60%	500x1500	very_rectangular	sap	jv	2.97x
metric_uniform	60%	500x500	default	jv	auction_scaled	1.51x
metric_uniform	60%	500x5000	very_rectangular	sap	hungarian	3.09x
tied	4 blocks	500x1500	sparse	lapmod	network_simplex	1.91x
tied	4 blocks	500x500	sparse	lapmod	auction_scaled	4.61x
tied	4 blocks	500x5000	sparse	lapmod	sap	2.09x
tied	complete	500x1500	very_rectangular	sap	network_simplex	13.38x

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
tied	complete	500x500	default	jv	network_simplex	10.28x
tied	complete	500x5000	very_rectangular	sap	network_simplex	7.29x
tied	25%	500x1500	sparse	lapmod	network_simplex	6.29x
tied	25%	500x500	sparse	lapmod	auction_scaled	12.44x
tied	25%	500x5000	sparse	lapmod	network_simplex	3.42x
tied	60%	500x1500	very_rectangular	sap	network_simplex	5.49x
tied	60%	500x500	default	jv	network_simplex	8.28x
tied	60%	500x5000	very_rectangular	sap	network_simplex	3.35x
binary	complete	1000x10000	no_cost_scale	hk01	jv	0.03x
binary	complete	1500x1500	no_cost_scale	hk01	jv	0.02x
binary	complete	1500x4500	no_cost_scale	hk01	jv	0.02x
binary	25%	1000x10000	no_cost_scale	hk01	sap	0.07x
binary	25%	1500x1500	no_cost_scale	hk01	sap	0.03x
binary	25%	1500x4500	no_cost_scale	hk01	sap	0.05x
dbl_uniform	complete	1000x10000	very_rectangular	sap	jv	6.16x
dbl_uniform	complete	1500x1500	default	jv	jv	0.99x
dbl_uniform	complete	1500x4500	very_rectangular	sap	jv	5.43x
dbl_uniform	25%	1000x10000	sparse	lapmod	jv	1.14x
dbl_uniform	25%	1500x1500	sparse	lapmod	jv	1.21x
dbl_uniform	25%	1500x4500	sparse	lapmod	lapmod	0.95x
heavy_tailed	complete	1000x10000	very_rectangular	sap	jv	6.00x
heavy_tailed	complete	1500x1500	default	jv	jv	1.00x
heavy_tailed	complete	1500x4500	very_rectangular	sap	jv	5.08x
heavy_tailed	25%	1000x10000	sparse	lapmod	jv	1.11x
heavy_tailed	25%	1500x1500	sparse	lapmod	jv	1.16x
heavy_tailed	25%	1500x4500	sparse	lapmod	lapmod	1.00x
int_uniform	complete	1000x10000	very_rectangular	sap	jv	6.02x
int_uniform	complete	1500x1500	default	jv	jv	1.00x
int_uniform	complete	1500x4500	very_rectangular	sap	jv	5.27x
int_uniform	25%	1000x10000	sparse	lapmod	jv	1.12x
int_uniform	25%	1500x1500	sparse	lapmod	jv	1.24x
int_uniform	25%	1500x4500	sparse	lapmod	jv	1.12x
metric_clustered	complete	1000x10000	very_rectangular	sap	jv	6.02x
metric_clustered	complete	1500x1500	default	jv	jv	1.00x
metric_clustered	complete	1500x4500	very_rectangular	sap	jv	5.34x
metric_clustered	25%	1000x10000	sparse	lapmod	jv	1.13x
metric_clustered	25%	1500x1500	sparse	lapmod	sap	1.23x
metric_clustered	25%	1500x4500	sparse	lapmod	jv	1.04x

2.3 Every solver, every cell

Table 6: Milliseconds per solver and cell: the median across instances, with the quartiles beside it. An instance contributes the median of its own repetitions, so the quartiles are between problems rather than between repetitions of one problem. A solver that errored or timed out on a cell has no row there.

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	4 blocks	500x500	hk01	3	1.66	1.65	1.67
binary	4 blocks	500x500	auto	3	1.68	1.68	1.69

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	4 blocks	500x500	network_simplex	3	4.31	4.14	4.65
binary	4 blocks	500x500	auction_scaled	3	4.64	4.63	4.66
binary	4 blocks	500x500	csa	3	4.8	4.77	4.8
binary	4 blocks	500x500	sap	3	11.2	11.2	11.3
binary	4 blocks	500x500	csflow	3	11.3	11.2	11.3
binary	4 blocks	500x500	ramshaw_tarjan	3	13.7	13.7	13.7
binary	4 blocks	500x500	hungarian	3	17.2	17.2	17.2
binary	4 blocks	500x500	jv	3	17.2	17.2	17.3
binary	4 blocks	500x500	lapmod	3	18.5	18.5	18.5
binary	4 blocks	500x500	sap_dense	3	25	24.9	25
binary	4 blocks	500x1500	hk01	3	4.68	4.68	4.7
binary	4 blocks	500x1500	auto	3	4.68	4.67	4.7
binary	4 blocks	500x1500	sap	3	26.1	26.1	26.2
binary	4 blocks	500x1500	csflow	3	26.2	26.1	26.2
binary	4 blocks	500x1500	network_simplex	3	30.1	28.3	30.7
binary	4 blocks	500x1500	ramshaw_tarjan	3	39.3	39.2	39.4
binary	4 blocks	500x1500	hungarian	3	52.2	52.2	52.3
binary	4 blocks	500x1500	jv	3	52.3	52.3	52.3
binary	4 blocks	500x1500	lapmod	3	56.6	56.6	56.6
binary	4 blocks	500x1500	sap_dense	3	74.9	74.9	75.3
binary	4 blocks	500x1500	csa	3	102	97	102
binary	4 blocks	500x1500	auction_scaled	3	160	158	163
binary	4 blocks	500x5000	auto	3	15.9	15.9	15.9
binary	4 blocks	500x5000	hk01	3	16	15.9	16
binary	4 blocks	500x5000	sap	3	77.5	77.5	77.6
binary	4 blocks	500x5000	csflow	3	77.6	77.6	77.7
binary	4 blocks	500x5000	ramshaw_tarjan	3	127	127	127
binary	4 blocks	500x5000	network_simplex	3	161	161	169
binary	4 blocks	500x5000	hungarian	3	175	175	175
binary	4 blocks	500x5000	jv	3	175	175	175
binary	4 blocks	500x5000	lapmod	3	189	189	189
binary	4 blocks	500x5000	sap_dense	3	250	250	250
binary	4 blocks	500x5000	csa	3	1.37e+03	1.36e+03	1.39e+03
binary	4 blocks	500x5000	auction_scaled	3	2.08e+03	2.08e+03	2.16e+03
binary	complete	500x500	hk01	3	3.45	3.45	3.47
binary	complete	500x500	auto	3	3.49	3.46	3.5
binary	complete	500x500	network_simplex	3	3.84	3.04	3.87
binary	complete	500x500	csa	3	14.1	14	14.1
binary	complete	500x500	auction_scaled	3	14.3	14.2	14.3
binary	complete	500x500	jv	3	58.4	57.2	58.6
binary	complete	500x500	hungarian	3	58.7	58.2	59.2
binary	complete	500x500	ramshaw_tarjan	3	63.1	63	63.2
binary	complete	500x500	sap_dense	3	87.2	86.9	88.4
binary	complete	500x500	lapmod	3	95.1	95	95.4
binary	complete	500x500	sap	3	106	106	106
binary	complete	500x500	csflow	3	107	107	107
binary	complete	500x1500	hk01	3	10	9.92	10
binary	complete	500x1500	auto	3	10	9.97	10
binary	complete	500x1500	network_simplex	3	21.8	20.9	26.8
binary	complete	500x1500	csa	3	87.5	82.6	87.6
binary	complete	500x1500	auction_scaled	3	128	127	128

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	complete	500x1500	jv	3	193	193	194
binary	complete	500x1500	hungarian	3	194	194	195
binary	complete	500x1500	ramshaw_tarjan	3	211	211	211
binary	complete	500x1500	sap	3	254	254	255
binary	complete	500x1500	csflow	3	257	257	257
binary	complete	500x1500	sap_dense	3	282	282	282
binary	complete	500x1500	lapmod	3	294	293	294
binary	complete	500x5000	auto	3	33.7	33.7	33.8
binary	complete	500x5000	hk01	3	33.8	33.7	33.8
binary	complete	500x5000	network_simplex	3	96.4	88.9	101
binary	complete	500x5000	jv	3	665	665	665
binary	complete	500x5000	hungarian	3	666	665	666
binary	complete	500x5000	ramshaw_tarjan	3	764	764	775
binary	complete	500x5000	sap	3	803	803	803
binary	complete	500x5000	csflow	3	812	812	812
binary	complete	500x5000	sap_dense	3	959	959	962
binary	complete	500x5000	lapmod	3	1.01e+03	1.01e+03	1.01e+03
binary	complete	500x5000	csa	3	1.1e+03	1.07e+03	1.24e+03
binary	complete	500x5000	auction_scaled	3	1.69e+03	1.69e+03	1.74e+03
binary	25%	500x500	auto	3	2.74	2.74	2.75
binary	25%	500x500	hk01	3	2.75	2.75	2.78
binary	25%	500x500	auction_scaled	3	6.13	6.12	6.16
binary	25%	500x500	network_simplex	3	6.35	6.19	6.38
binary	25%	500x500	csa	3	6.41	6.4	6.43
binary	25%	500x500	sap	3	45.1	45	45.1
binary	25%	500x500	csflow	3	45.2	45.1	45.2
binary	25%	500x500	jv	3	61.4	61.1	61.8
binary	25%	500x500	hungarian	3	61.6	61.5	61.8
binary	25%	500x500	lapmod	3	71.4	71.4	71.5
binary	25%	500x500	ramshaw_tarjan	3	99.4	99.4	99.7
binary	25%	500x500	sap_dense	3	127	127	127
binary	25%	500x1500	hk01	3	7.95	7.94	7.96
binary	25%	500x1500	auto	3	7.96	7.93	7.96
binary	25%	500x1500	network_simplex	3	30.1	29.6	30.3
binary	25%	500x1500	csa	3	84.4	79.6	84.4
binary	25%	500x1500	sap	3	100	99.9	100
binary	25%	500x1500	csflow	3	101	100	101
binary	25%	500x1500	auction_scaled	3	109	106	115
binary	25%	500x1500	jv	3	199	199	199
binary	25%	500x1500	hungarian	3	200	199	200
binary	25%	500x1500	lapmod	3	220	220	221
binary	25%	500x1500	ramshaw_tarjan	3	385	385	385
binary	25%	500x1500	sap_dense	3	478	478	479
binary	25%	500x5000	hk01	3	26.6	26.5	26.6
binary	25%	500x5000	auto	3	26.6	26.5	26.6
binary	25%	500x5000	network_simplex	3	169	166	174
binary	25%	500x5000	sap	3	296	295	296
binary	25%	500x5000	csflow	3	299	298	300
binary	25%	500x5000	hungarian	3	683	683	684
binary	25%	500x5000	jv	3	683	683	684
binary	25%	500x5000	lapmod	3	740	740	741

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	25%	500x5000	csa	3	1.11e+03	1.11e+03	1.23e+03
binary	25%	500x5000	ramshaw_tarjan	3	1.37e+03	1.37e+03	1.37e+03
binary	25%	500x5000	auction_scaled	3	1.67e+03	1.63e+03	1.69e+03
binary	25%	500x5000	sap_dense	3	1.7e+03	1.7e+03	1.7e+03
binary	60%	500x500	hk01	3	4.25	4.24	4.26
binary	60%	500x500	auto	3	4.32	4.27	4.34
binary	60%	500x500	network_simplex	3	6.15	5.92	6.2
binary	60%	500x500	auction_scaled	3	11.3	11.2	11.3
binary	60%	500x500	csa	3	11.4	11.4	11.5
binary	60%	500x500	hungarian	3	61.3	61	61.7
binary	60%	500x500	jv	3	61.4	61.3	62.4
binary	60%	500x500	sap	3	75.1	74.9	75.1
binary	60%	500x500	csflow	3	75.5	75.4	75.6
binary	60%	500x500	lapmod	3	85.4	85.4	85.6
binary	60%	500x500	ramshaw_tarjan	3	140	140	140
binary	60%	500x500	sap_dense	3	142	142	142
binary	60%	500x1500	hk01	3	12.2	12.2	12.4
binary	60%	500x1500	auto	3	12.3	12.3	12.3
binary	60%	500x1500	network_simplex	3	35.6	33.9	37.1
binary	60%	500x1500	csa	3	95.3	92.9	95.5
binary	60%	500x1500	auction_scaled	3	126	124	137
binary	60%	500x1500	sap	3	177	176	177
binary	60%	500x1500	csflow	3	178	178	178
binary	60%	500x1500	hungarian	3	200	200	201
binary	60%	500x1500	jv	3	201	201	201
binary	60%	500x1500	lapmod	3	260	260	261
binary	60%	500x1500	ramshaw_tarjan	3	542	541	543
binary	60%	500x1500	sap_dense	3	592	591	592
binary	60%	500x5000	auto	3	41.6	41.5	41.6
binary	60%	500x5000	hk01	3	41.6	41.5	41.7
binary	60%	500x5000	network_simplex	3	167	163	170
binary	60%	500x5000	sap	3	552	552	553
binary	60%	500x5000	csflow	3	558	558	558
binary	60%	500x5000	jv	3	686	686	687
binary	60%	500x5000	hungarian	3	687	686	687
binary	60%	500x5000	lapmod	3	874	874	875
binary	60%	500x5000	csa	3	1.3e+03	1.21e+03	1.33e+03
binary	60%	500x5000	auction_scaled	3	1.65e+03	1.59e+03	1.66e+03
binary	60%	500x5000	ramshaw_tarjan	3	1.96e+03	1.95e+03	1.96e+03
binary	60%	500x5000	sap_dense	3	2.13e+03	2.13e+03	2.13e+03
constant	4 blocks	500x500	hk01	3	1.25	1.25	1.27
constant	4 blocks	500x500	auto	3	1.25	1.25	1.26
constant	4 blocks	500x500	network_simplex	3	2.48	2.47	2.53
constant	4 blocks	500x500	auction_scaled	3	3.1	3.1	3.12
constant	4 blocks	500x500	csa	3	3.16	3.16	3.2
constant	4 blocks	500x500	sap	3	8.78	8.77	8.8
constant	4 blocks	500x500	csflow	3	8.81	8.79	8.85
constant	4 blocks	500x500	ramshaw_tarjan	3	12.5	12.5	12.5
constant	4 blocks	500x500	hungarian	3	16.8	16.8	16.9
constant	4 blocks	500x500	jv	3	16.9	16.9	16.9
constant	4 blocks	500x500	lapmod	3	17.7	17.7	17.7

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	4 blocks	500x500	sap_dense	3	24.7	24.6	24.7
constant	4 blocks	500x1500	auto	3	3.55	3.54	3.55
constant	4 blocks	500x1500	hk01	3	3.55	3.54	3.55
constant	4 blocks	500x1500	network_simplex	3	16.4	16.4	16.4
constant	4 blocks	500x1500	csflow	3	21.9	21.8	21.9
constant	4 blocks	500x1500	sap	3	21.9	21.8	21.9
constant	4 blocks	500x1500	ramshaw_tarjan	3	36.3	36.3	36.4
constant	4 blocks	500x1500	hungarian	3	51.3	51.3	51.3
constant	4 blocks	500x1500	jv	3	51.3	51.3	51.3
constant	4 blocks	500x1500	lapmod	3	55.7	55.7	55.7
constant	4 blocks	500x1500	sap_dense	3	74	74	74.1
constant	4 blocks	500x1500	csa	3	101	101	101
constant	4 blocks	500x1500	auction_scaled	3	149	149	149
constant	4 blocks	500x5000	auto	3	12	12	12.1
constant	4 blocks	500x5000	hk01	3	12.1	12	12.1
constant	4 blocks	500x5000	sap	3	66.5	66.5	66.5
constant	4 blocks	500x5000	csflow	3	67	66.9	67
constant	4 blocks	500x5000	network_simplex	3	93.2	93.1	93.3
constant	4 blocks	500x5000	ramshaw_tarjan	3	119	119	119
constant	4 blocks	500x5000	hungarian	3	173	172	173
constant	4 blocks	500x5000	jv	3	173	173	173
constant	4 blocks	500x5000	lapmod	3	186	186	186
constant	4 blocks	500x5000	sap_dense	3	247	247	247
constant	4 blocks	500x5000	csa	3	1.3e+03	1.3e+03	1.3e+03
constant	4 blocks	500x5000	auction_scaled	3	2.47e+03	2.47e+03	2.47e+03
constant	complete	500x500	hk01	3	1.74	1.74	1.74
constant	complete	500x500	auto	3	1.74	1.74	1.75
constant	complete	500x500	network_simplex	3	2.3	2.29	2.37
constant	complete	500x500	csa	3	8.87	8.86	8.87
constant	complete	500x500	auction_scaled	3	9.19	9.19	9.2
constant	complete	500x500	jv	3	56.2	56.2	57
constant	complete	500x500	hungarian	3	56.7	56.3	57.4
constant	complete	500x500	ramshaw_tarjan	3	58.9	58.7	59
constant	complete	500x500	sap_dense	3	85.5	85.3	86.1
constant	complete	500x500	lapmod	3	91.5	91.1	91.5
constant	complete	500x500	sap	3	97.5	97.4	97.5
constant	complete	500x500	csflow	3	98.3	98.3	98.5
constant	complete	500x1500	hk01	3	4.96	4.93	4.99
constant	complete	500x1500	auto	3	5	4.97	5.05
constant	complete	500x1500	network_simplex	3	7.85	7.8	7.9
constant	complete	500x1500	csa	3	77.3	76.5	77.3
constant	complete	500x1500	auction_scaled	3	184	184	184
constant	complete	500x1500	hungarian	3	190	190	191
constant	complete	500x1500	jv	3	190	190	191
constant	complete	500x1500	ramshaw_tarjan	3	202	202	203
constant	complete	500x1500	sap	3	237	237	238
constant	complete	500x1500	csflow	3	240	240	240
constant	complete	500x1500	sap_dense	3	278	278	278
constant	complete	500x1500	lapmod	3	287	287	288
constant	complete	500x5000	hk01	3	16.5	16.5	16.6
constant	complete	500x5000	auto	3	16.6	16.6	16.7

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	complete	500x5000	network_simplex	3	27.6	27.6	27.8
constant	complete	500x5000	hungarian	3	653	653	654
constant	complete	500x5000	jv	3	654	653	654
constant	complete	500x5000	ramshaw_tarjan	3	738	736	746
constant	complete	500x5000	sap	3	757	757	777
constant	complete	500x5000	csflow	3	765	765	766
constant	complete	500x5000	csa	3	926	926	926
constant	complete	500x5000	sap_dense	3	948	947	951
constant	complete	500x5000	lapmod	3	980	980	981
constant	complete	500x5000	auction_scaled	3	2.31e+03	2.31e+03	2.32e+03
constant	25%	500x500	hk01	3	2.35	2.34	2.36
constant	25%	500x500	auto	3	2.36	2.36	2.38
constant	25%	500x500	network_simplex	3	3.66	3.57	3.66
constant	25%	500x500	auction_scaled	3	5.59	5.33	5.6
constant	25%	500x500	csa	3	5.75	5.52	5.77
constant	25%	500x500	sap	3	37	36.9	37
constant	25%	500x500	csflow	3	37.1	37.1	37.1
constant	25%	500x500	hungarian	3	60.6	59.9	61.1
constant	25%	500x500	jv	3	60.8	60.8	61.3
constant	25%	500x500	lapmod	3	69.9	69.8	69.9
constant	25%	500x500	ramshaw_tarjan	3	93.6	93.2	93.8
constant	25%	500x500	sap_dense	3	127	126	127
constant	25%	500x1500	auto	3	6.66	6.63	6.73
constant	25%	500x1500	hk01	3	6.67	6.65	6.7
constant	25%	500x1500	network_simplex	3	23.6	23.4	24.6
constant	25%	500x1500	csa	3	65.7	64.4	65.9
constant	25%	500x1500	sap	3	82.4	82.4	82.8
constant	25%	500x1500	csflow	3	83.1	83	83.5
constant	25%	500x1500	auction_scaled	3	128	126	133
constant	25%	500x1500	hungarian	3	198	198	198
constant	25%	500x1500	jv	3	199	198	199
constant	25%	500x1500	lapmod	3	218	218	218
constant	25%	500x1500	ramshaw_tarjan	3	373	372	373
constant	25%	500x1500	sap_dense	3	477	476	477
constant	25%	500x5000	auto	3	22	22	22.2
constant	25%	500x5000	hk01	3	22.1	22.1	22.2
constant	25%	500x5000	network_simplex	3	116	115	116
constant	25%	500x5000	csflow	3	250	249	250
constant	25%	500x5000	sap	3	250	248	250
constant	25%	500x5000	hungarian	3	676	676	677
constant	25%	500x5000	jv	3	677	677	677
constant	25%	500x5000	lapmod	3	732	732	732
constant	25%	500x5000	csa	3	968	887	1e+03
constant	25%	500x5000	ramshaw_tarjan	3	1.34e+03	1.33e+03	1.34e+03
constant	25%	500x5000	sap_dense	3	1.69e+03	1.69e+03	1.69e+03
constant	25%	500x5000	auction_scaled	3	1.88e+03	1.87e+03	2.03e+03
constant	60%	500x500	auto	3	3.22	3.22	3.27
constant	60%	500x500	hk01	3	3.23	3.22	3.24
constant	60%	500x500	network_simplex	3	4.18	4.13	4.21
constant	60%	500x500	auction_scaled	3	8.94	8.5	9.39
constant	60%	500x500	csa	3	9.14	8.71	9.58

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	60%	500x500	hungarian	3	61.4	61	61.4
constant	60%	500x500	jv	3	62	61.3	62.1
constant	60%	500x500	sap	3	67.6	67.6	67.7
constant	60%	500x500	csflow	3	68	68	68.1
constant	60%	500x500	lapmod	3	82.2	81.7	82.3
constant	60%	500x500	ramshaw_tarjan	3	134	134	134
constant	60%	500x500	sap_dense	3	140	140	140
constant	60%	500x1500	hk01	3	9.12	9.12	9.25
constant	60%	500x1500	auto	3	9.16	9.1	9.19
constant	60%	500x1500	network_simplex	3	27.3	24.8	27.5
constant	60%	500x1500	csa	3	83.6	78.7	83.6
constant	60%	500x1500	sap	3	160	160	160
constant	60%	500x1500	csflow	3	162	162	162
constant	60%	500x1500	auction_scaled	3	164	138	171
constant	60%	500x1500	hungarian	3	197	196	198
constant	60%	500x1500	jv	3	198	198	199
constant	60%	500x1500	lapmod	3	256	256	256
constant	60%	500x1500	ramshaw_tarjan	3	531	530	531
constant	60%	500x1500	sap_dense	3	591	590	591
constant	60%	500x5000	auto	3	30.5	30.5	30.9
constant	60%	500x5000	hk01	3	30.5	30.4	30.8
constant	60%	500x5000	network_simplex	3	102	98.7	108
constant	60%	500x5000	csflow	3	510	510	525
constant	60%	500x5000	sap	3	537	521	537
constant	60%	500x5000	hungarian	3	676	676	676
constant	60%	500x5000	jv	3	677	677	677
constant	60%	500x5000	lapmod	3	861	861	861
constant	60%	500x5000	csa	3	1.04e+03	995	1.09e+03
constant	60%	500x5000	ramshaw_tarjan	3	1.9e+03	1.9e+03	1.91e+03
constant	60%	500x5000	sap_dense	3	2.12e+03	2.12e+03	2.12e+03
constant	60%	500x5000	auction_scaled	3	2.16e+03	2.05e+03	2.25e+03
dbl_uniform	4 blocks	500x500	auction_scaled	3	2.64	2.62	2.64
dbl_uniform	4 blocks	500x500	csa	3	3.37	3.36	3.4
dbl_uniform	4 blocks	500x500	hungarian	3	3.63	3.58	3.67
dbl_uniform	4 blocks	500x500	jv	3	3.68	3.6	3.7
dbl_uniform	4 blocks	500x500	lapmod	3	4.21	4.14	4.25
dbl_uniform	4 blocks	500x500	auto	3	4.22	4.14	4.25
dbl_uniform	4 blocks	500x500	ramshaw_tarjan	3	4.94	4.88	4.96
dbl_uniform	4 blocks	500x500	csflow	3	4.97	4.93	5
dbl_uniform	4 blocks	500x500	sap	3	5.03	4.99	5.1
dbl_uniform	4 blocks	500x500	sap_dense	3	5.48	5.36	5.5
dbl_uniform	4 blocks	500x500	network_simplex	3	27.9	27.3	28.2
dbl_uniform	4 blocks	500x1500	hungarian	3	3.67	3.66	3.72
dbl_uniform	4 blocks	500x1500	jv	3	3.67	3.67	3.73
dbl_uniform	4 blocks	500x1500	auto	3	4.28	4.26	4.35
dbl_uniform	4 blocks	500x1500	lapmod	3	4.39	4.36	4.4
dbl_uniform	4 blocks	500x1500	sap_dense	3	6.2	6.16	6.26
dbl_uniform	4 blocks	500x1500	ramshaw_tarjan	3	7.61	7.52	7.66
dbl_uniform	4 blocks	500x1500	csflow	3	9.45	9.34	9.46
dbl_uniform	4 blocks	500x1500	sap	3	9.69	9.57	9.76
dbl_uniform	4 blocks	500x1500	network_simplex	3	92.8	91.2	95.3

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	4 blocks	500x1500	auction_scaled	3	281	268	286
dbl_uniform	4 blocks	500x1500	csa	3	506	498	564
dbl_uniform	4 blocks	500x5000	hungarian	3	11.9	11.9	11.9
dbl_uniform	4 blocks	500x5000	jv	3	12	11.9	12
dbl_uniform	4 blocks	500x5000	auto	3	13.5	13.4	13.6
dbl_uniform	4 blocks	500x5000	lapmod	3	13.5	13.5	13.5
dbl_uniform	4 blocks	500x5000	sap_dense	3	19	19	19.1
dbl_uniform	4 blocks	500x5000	ramshaw_tarjan	3	23.8	23.7	23.8
dbl_uniform	4 blocks	500x5000	csflow	3	30.3	30.3	30.6
dbl_uniform	4 blocks	500x5000	sap	3	31.1	30.9	31.1
dbl_uniform	4 blocks	500x5000	network_simplex	3	527	518	534
dbl_uniform	4 blocks	500x5000	auction_scaled	3	3.75e+03	3.69e+03	3.78e+03
dbl_uniform	4 blocks	500x5000	csa	3	1.08e+04	1.07e+04	1.11e+04
dbl_uniform	complete	500x500	jv	3	3.19	3.17	3.38
dbl_uniform	complete	500x500	auto	3	3.23	3.18	3.4
dbl_uniform	complete	500x500	hungarian	3	7.7	7.63	7.96
dbl_uniform	complete	500x500	auction_scaled	3	8.14	8.07	8.25
dbl_uniform	complete	500x500	sap_dense	3	10.3	10.2	10.7
dbl_uniform	complete	500x500	csa	3	10.3	10.2	10.4
dbl_uniform	complete	500x500	lapmod	3	10.7	10.6	11.1
dbl_uniform	complete	500x500	ramshaw_tarjan	3	13.4	13.3	13.7
dbl_uniform	complete	500x500	sap	3	19.4	19.3	19.8
dbl_uniform	complete	500x500	csflow	3	19.4	19.3	19.7
dbl_uniform	complete	500x500	network_simplex	3	91.5	85.5	98
dbl_uniform	complete	500x1500	hungarian	3	3.93	3.86	4.01
dbl_uniform	complete	500x1500	jv	3	3.94	3.84	4.04
dbl_uniform	complete	500x1500	lapmod	3	5.41	5.28	5.45
dbl_uniform	complete	500x1500	sap_dense	3	6.17	6	6.21
dbl_uniform	complete	500x1500	ramshaw_tarjan	3	14.2	14	14.4
dbl_uniform	complete	500x1500	auto	3	26.8	26.7	27
dbl_uniform	complete	500x1500	csflow	3	26.9	26.9	27.2
dbl_uniform	complete	500x1500	sap	3	27.1	27	27.1
dbl_uniform	complete	500x1500	network_simplex	3	95.3	94.4	120
dbl_uniform	complete	500x1500	auction_scaled	3	266	251	273
dbl_uniform	complete	500x1500	csa	3	507	493	544
dbl_uniform	complete	500x5000	jv	3	11.4	11.3	11.5
dbl_uniform	complete	500x5000	hungarian	3	11.5	11.3	11.6
dbl_uniform	complete	500x5000	lapmod	3	16.4	16.3	16.4
dbl_uniform	complete	500x5000	sap_dense	3	18.1	18	18.2
dbl_uniform	complete	500x5000	ramshaw_tarjan	3	46.4	46.1	46.4
dbl_uniform	complete	500x5000	sap	3	100	100	101
dbl_uniform	complete	500x5000	csflow	3	100	100	101
dbl_uniform	complete	500x5000	auto	3	102	101	102
dbl_uniform	complete	500x5000	network_simplex	3	544	543	568
dbl_uniform	complete	500x5000	auction_scaled	3	3.23e+03	3.22e+03	3.27e+03
dbl_uniform	complete	500x5000	csa	3	1.01e+04	9.89e+03	1.06e+04
dbl_uniform	25%	500x500	auction_scaled	3	4.2	4.16	4.27
dbl_uniform	25%	500x500	csa	3	5.41	5.38	5.41
dbl_uniform	25%	500x500	jv	3	8.55	8.22	9.05
dbl_uniform	25%	500x500	hungarian	3	8.56	8.22	9.05
dbl_uniform	25%	500x500	csflow	3	9.91	9.77	10.4

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	25%	500x500	sap	3	9.93	9.73	10.4
dbl_uniform	25%	500x500	lapmod	3	10.1	9.63	10.8
dbl_uniform	25%	500x500	auto	3	10.2	9.66	10.8
dbl_uniform	25%	500x500	sap_dense	3	17.4	16.5	18.5
dbl_uniform	25%	500x500	ramshaw_tarjan	3	17.6	16.8	18.8
dbl_uniform	25%	500x500	network_simplex	3	85.8	81.8	91.2
dbl_uniform	25%	500x1500	jv	3	6.18	6.18	6.22
dbl_uniform	25%	500x1500	hungarian	3	6.2	6.18	6.21
dbl_uniform	25%	500x1500	lapmod	3	7.09	7.09	7.11
dbl_uniform	25%	500x1500	auto	3	7.14	7.1	7.15
dbl_uniform	25%	500x1500	sap_dense	3	10.1	10.1	10.2
dbl_uniform	25%	500x1500	ramshaw_tarjan	3	11.3	11.3	11.4
dbl_uniform	25%	500x1500	csflow	3	12.5	12.5	12.6
dbl_uniform	25%	500x1500	sap	3	12.6	12.6	12.6
dbl_uniform	25%	500x1500	network_simplex	3	109	104	154
dbl_uniform	25%	500x1500	auction_scaled	3	239	238	242
dbl_uniform	25%	500x1500	csa	3	577	528	606
dbl_uniform	25%	500x5000	hungarian	3	19.6	19.5	19.6
dbl_uniform	25%	500x5000	jv	3	19.8	19.7	20.2
dbl_uniform	25%	500x5000	auto	3	23.1	23.1	23.2
dbl_uniform	25%	500x5000	lapmod	3	23.2	23.1	23.2
dbl_uniform	25%	500x5000	sap_dense	3	32	31.9	32.4
dbl_uniform	25%	500x5000	ramshaw_tarjan	3	36	35.9	36.1
dbl_uniform	25%	500x5000	csflow	3	41	41	41.1
dbl_uniform	25%	500x5000	sap	3	41.6	41.6	41.8
dbl_uniform	25%	500x5000	network_simplex	3	575	574	595
dbl_uniform	25%	500x5000	auction_scaled	3	3.4e+03	3.4e+03	3.44e+03
dbl_uniform	25%	500x5000	csa	3	1.03e+04	1e+04	1.09e+04
dbl_uniform	60%	500x500	auction_scaled	3	7.48	7.33	7.57
dbl_uniform	60%	500x500	auto	3	8.94	8.58	9.64
dbl_uniform	60%	500x500	hungarian	3	8.96	8.58	9.64
dbl_uniform	60%	500x500	jv	3	8.97	8.58	9.64
dbl_uniform	60%	500x500	csa	3	9.73	9.6	9.81
dbl_uniform	60%	500x500	lapmod	3	12.3	11.8	13.8
dbl_uniform	60%	500x500	csflow	3	16.3	15.8	17
dbl_uniform	60%	500x500	sap	3	16.4	15.9	17
dbl_uniform	60%	500x500	sap_dense	3	19.2	18.3	21.1
dbl_uniform	60%	500x500	ramshaw_tarjan	3	22.1	21.1	23.9
dbl_uniform	60%	500x500	network_simplex	3	98.8	87	99.4
dbl_uniform	60%	500x1500	jv	3	7.92	7.89	8.02
dbl_uniform	60%	500x1500	hungarian	3	7.98	7.93	8.01
dbl_uniform	60%	500x1500	lapmod	3	9.58	9.56	9.64
dbl_uniform	60%	500x1500	sap_dense	3	13.1	13	13.1
dbl_uniform	60%	500x1500	ramshaw_tarjan	3	17.5	17.3	17.7
dbl_uniform	60%	500x1500	csflow	3	22.9	22.9	23.2
dbl_uniform	60%	500x1500	auto	3	23.3	23.1	23.5
dbl_uniform	60%	500x1500	sap	3	23.5	23.2	23.6
dbl_uniform	60%	500x1500	network_simplex	3	117	116	129
dbl_uniform	60%	500x1500	auction_scaled	3	271	267	285
dbl_uniform	60%	500x1500	csa	3	579	550	779
dbl_uniform	60%	500x5000	jv	3	24.6	24.5	24.8

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	60%	500x5000	hungarian	3	24.6	24.5	24.8
dbl_uniform	60%	500x5000	lapmod	3	30.8	30.7	30.8
dbl_uniform	60%	500x5000	sap_dense	3	41.3	41.2	41.4
dbl_uniform	60%	500x5000	ramshaw_tarjan	3	54.7	54.6	55
dbl_uniform	60%	500x5000	csflow	3	74.8	74.7	74.8
dbl_uniform	60%	500x5000	sap	3	75.2	74.7	75.2
dbl_uniform	60%	500x5000	auto	3	75.2	74.7	75.3
dbl_uniform	60%	500x5000	network_simplex	3	684	645	697
dbl_uniform	60%	500x5000	auction_scaled	3	3.19e+03	3e+03	3.33e+03
dbl_uniform	60%	500x5000	csa	3	1e+04	1e+04	1.02e+04
heavy_tailed	4 blocks	500x500	hungarian	3	4.18	3.91	4.29
heavy_tailed	4 blocks	500x500	jv	3	4.18	3.92	4.31
heavy_tailed	4 blocks	500x500	auto	3	4.87	4.56	5
heavy_tailed	4 blocks	500x500	lapmod	3	4.87	4.56	5
heavy_tailed	4 blocks	500x500	csflow	3	5.3	5.12	5.37
heavy_tailed	4 blocks	500x500	sap	3	5.33	5.16	5.46
heavy_tailed	4 blocks	500x500	ramshaw_tarjan	3	5.42	5.23	5.51
heavy_tailed	4 blocks	500x500	auction_scaled	3	5.82	5.78	5.87
heavy_tailed	4 blocks	500x500	sap_dense	3	6.19	5.84	6.35
heavy_tailed	4 blocks	500x500	csa	3	7	6.83	7.06
heavy_tailed	4 blocks	500x500	network_simplex	3	28.9	27.3	29.9
heavy_tailed	4 blocks	500x1500	jv	3	3.66	3.66	3.66
heavy_tailed	4 blocks	500x1500	hungarian	3	3.69	3.67	3.76
heavy_tailed	4 blocks	500x1500	lapmod	3	4.21	4.21	4.26
heavy_tailed	4 blocks	500x1500	auto	3	4.24	4.22	4.3
heavy_tailed	4 blocks	500x1500	sap_dense	3	6.06	6.06	6.14
heavy_tailed	4 blocks	500x1500	ramshaw_tarjan	3	7.44	7.41	7.45
heavy_tailed	4 blocks	500x1500	csflow	3	9.4	9.32	9.42
heavy_tailed	4 blocks	500x1500	sap	3	9.59	9.49	9.59
heavy_tailed	4 blocks	500x1500	network_simplex	3	109	103	110
heavy_tailed	4 blocks	500x1500	csa	3	184	182	203
heavy_tailed	4 blocks	500x1500	auction_scaled	1	264	264	264
heavy_tailed	4 blocks	500x5000	jv	3	12	12	12
heavy_tailed	4 blocks	500x5000	lapmod	3	13.5	13.5	13.6
heavy_tailed	4 blocks	500x5000	auto	3	13.6	13.6	18
heavy_tailed	4 blocks	500x5000	hungarian	3	18.8	15.4	18.8
heavy_tailed	4 blocks	500x5000	ramshaw_tarjan	3	23.8	23.7	27.2
heavy_tailed	4 blocks	500x5000	csflow	3	30.5	30.5	31
heavy_tailed	4 blocks	500x5000	sap_dense	3	30.7	30.1	30.8
heavy_tailed	4 blocks	500x5000	sap	3	39.1	38.4	39.5
heavy_tailed	4 blocks	500x5000	network_simplex	3	669	655	686
heavy_tailed	4 blocks	500x5000	csa	3	7.43e+03	7.38e+03	7.43e+03
heavy_tailed	complete	500x500	auto	3	3.62	3.61	3.81
heavy_tailed	complete	500x500	jv	3	3.64	3.63	3.84
heavy_tailed	complete	500x500	hungarian	3	8.11	7.62	8.67
heavy_tailed	complete	500x500	sap_dense	3	10.9	10.2	11.7
heavy_tailed	complete	500x500	lapmod	3	11.4	10.7	12.3
heavy_tailed	complete	500x500	ramshaw_tarjan	3	13.7	13.2	14.4
heavy_tailed	complete	500x500	csa	3	18.4	17.8	18.7
heavy_tailed	complete	500x500	csflow	3	19.6	19	20.5
heavy_tailed	complete	500x500	sap	3	19.7	19.1	20.5

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	complete	500x500	auction_scaled	3	20.6	20.2	21
heavy_tailed	complete	500x500	network_simplex	3	88.1	87	92.6
heavy_tailed	complete	500x1500	jv	3	3.74	3.72	3.78
heavy_tailed	complete	500x1500	hungarian	3	3.82	3.76	3.86
heavy_tailed	complete	500x1500	lapmod	3	5.28	5.26	5.47
heavy_tailed	complete	500x1500	sap_dense	3	5.9	5.87	5.96
heavy_tailed	complete	500x1500	ramshaw_tarjan	3	13.9	13.8	14
heavy_tailed	complete	500x1500	sap	3	26.5	26.4	26.6
heavy_tailed	complete	500x1500	auto	3	26.7	26.6	26.8
heavy_tailed	complete	500x1500	csflow	3	26.7	26.5	26.7
heavy_tailed	complete	500x1500	network_simplex	3	112	107	130
heavy_tailed	complete	500x1500	csa	3	148	147	150
heavy_tailed	complete	500x1500	auction_scaled	1	257	257	257
heavy_tailed	complete	500x5000	hungarian	3	11.2	11.1	14.9
heavy_tailed	complete	500x5000	jv	3	11.2	11.2	11.3
heavy_tailed	complete	500x5000	lapmod	3	16.6	16.4	21.4
heavy_tailed	complete	500x5000	sap_dense	3	22.4	20.2	25.7
heavy_tailed	complete	500x5000	ramshaw_tarjan	3	52.7	49.2	54.1
heavy_tailed	complete	500x5000	auto	3	100	100	105
heavy_tailed	complete	500x5000	csflow	3	101	100	102
heavy_tailed	complete	500x5000	sap	3	108	104	110
heavy_tailed	complete	500x5000	network_simplex	3	787	740	808
heavy_tailed	complete	500x5000	csa	3	7.08e+03	6.95e+03	7.09e+03
heavy_tailed	25%	500x500	auction_scaled	3	7.39	7.22	7.55
heavy_tailed	25%	500x500	csa	3	8.69	8.53	9.04
heavy_tailed	25%	500x500	jv	3	10.1	8.74	10.1
heavy_tailed	25%	500x500	hungarian	3	10.1	8.76	10.2
heavy_tailed	25%	500x500	csflow	3	10.8	9.99	11
heavy_tailed	25%	500x500	sap	3	10.8	10	11
heavy_tailed	25%	500x500	lapmod	3	12.1	10.3	12.2
heavy_tailed	25%	500x500	auto	3	12.1	10.3	12.2
heavy_tailed	25%	500x500	sap_dense	3	20.9	17.7	21
heavy_tailed	25%	500x500	ramshaw_tarjan	3	21	17.9	21
heavy_tailed	25%	500x500	network_simplex	3	93.8	89.2	95.2
heavy_tailed	25%	500x1500	jv	3	6.3	6.29	6.34
heavy_tailed	25%	500x1500	hungarian	3	6.31	6.25	6.33
heavy_tailed	25%	500x1500	lapmod	3	7.09	7.09	7.12
heavy_tailed	25%	500x1500	auto	3	7.13	7.1	7.17
heavy_tailed	25%	500x1500	sap_dense	3	10.2	10.2	10.2
heavy_tailed	25%	500x1500	ramshaw_tarjan	3	11.5	11.4	11.5
heavy_tailed	25%	500x1500	csflow	3	12.6	12.5	12.6
heavy_tailed	25%	500x1500	sap	3	12.7	12.7	12.8
heavy_tailed	25%	500x1500	csa	3	127	119	129
heavy_tailed	25%	500x1500	network_simplex	3	133	128	147
heavy_tailed	25%	500x1500	auction_scaled	3	231	227	248
heavy_tailed	25%	500x5000	hungarian	3	19.8	19.7	23.1
heavy_tailed	25%	500x5000	lapmod	3	23.2	23.1	23.2
heavy_tailed	25%	500x5000	auto	3	23.2	23.2	27.5
heavy_tailed	25%	500x5000	jv	3	26.5	23.1	27.5
heavy_tailed	25%	500x5000	ramshaw_tarjan	3	36	36	39.5
heavy_tailed	25%	500x5000	sap_dense	3	37.1	37.1	40.5

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	25%	500x5000	csflow	3	41.4	41.3	46.2
heavy_tailed	25%	500x5000	sap	3	42.3	41.9	46.1
heavy_tailed	25%	500x5000	network_simplex	3	698	684	752
heavy_tailed	25%	500x5000	csa	3	6.77e+03	6.67e+03	6.79e+03
heavy_tailed	60%	500x500	jv	3	8.58	8.53	9.18
heavy_tailed	60%	500x500	hungarian	3	8.61	8.56	9.18
heavy_tailed	60%	500x500	auto	3	8.61	8.55	9.19
heavy_tailed	60%	500x500	lapmod	3	12.2	12.1	13.3
heavy_tailed	60%	500x500	auction_scaled	3	14.2	14.2	14.3
heavy_tailed	60%	500x500	csa	3	14.3	14.1	14.5
heavy_tailed	60%	500x500	sap	3	15.8	15.8	16.5
heavy_tailed	60%	500x500	csflow	3	15.8	15.8	16.5
heavy_tailed	60%	500x500	sap_dense	3	18.4	18.2	20
heavy_tailed	60%	500x500	ramshaw_tarjan	3	20.9	20.8	22.6
heavy_tailed	60%	500x500	network_simplex	3	85.6	84.9	87.6
heavy_tailed	60%	500x1500	jv	3	7.9	7.83	7.9
heavy_tailed	60%	500x1500	hungarian	3	7.92	7.78	7.95
heavy_tailed	60%	500x1500	lapmod	3	9.52	9.48	9.55
heavy_tailed	60%	500x1500	sap_dense	3	12.8	12.8	13
heavy_tailed	60%	500x1500	ramshaw_tarjan	3	17.2	17.2	17.4
heavy_tailed	60%	500x1500	csflow	3	22.8	22.8	23
heavy_tailed	60%	500x1500	sap	3	23	23	23.1
heavy_tailed	60%	500x1500	auto	3	23.2	23	23.2
heavy_tailed	60%	500x1500	csa	3	132	127	138
heavy_tailed	60%	500x1500	network_simplex	3	168	159	185
heavy_tailed	60%	500x5000	jv	3	27.2	26.3	29.5
heavy_tailed	60%	500x5000	hungarian	3	27.5	27.2	29.6
heavy_tailed	60%	500x5000	lapmod	3	30.9	30.8	36.7
heavy_tailed	60%	500x5000	sap_dense	3	45.6	45.5	49.7
heavy_tailed	60%	500x5000	ramshaw_tarjan	3	55.2	55.1	58.9
heavy_tailed	60%	500x5000	csflow	3	77	76.1	77.1
heavy_tailed	60%	500x5000	sap	3	77.5	76.6	81.4
heavy_tailed	60%	500x5000	auto	3	83.8	79.3	84.8
heavy_tailed	60%	500x5000	network_simplex	3	740	726	751
heavy_tailed	60%	500x5000	csa	3	6.95e+03	6.92e+03	6.95e+03
int_uniform	4 blocks	500x500	auction_scaled	3	2.91	2.89	2.96
int_uniform	4 blocks	500x500	csa	3	3.07	3.04	3.12
int_uniform	4 blocks	500x500	jv	3	3.72	3.71	3.87
int_uniform	4 blocks	500x500	hungarian	3	3.76	3.74	3.91
int_uniform	4 blocks	500x500	lapmod	3	4.31	4.3	4.5
int_uniform	4 blocks	500x500	auto	3	4.32	4.31	4.51
int_uniform	4 blocks	500x500	ramshaw_tarjan	3	5.08	5.06	5.18
int_uniform	4 blocks	500x500	csflow	3	5.16	5.11	5.21
int_uniform	4 blocks	500x500	sap	3	5.29	5.2	5.3
int_uniform	4 blocks	500x500	sap_dense	3	5.59	5.56	5.83
int_uniform	4 blocks	500x500	network_simplex	3	28.4	27.4	29.5
int_uniform	4 blocks	500x1500	hungarian	3	3.74	3.68	3.74
int_uniform	4 blocks	500x1500	jv	3	3.76	3.67	3.8
int_uniform	4 blocks	500x1500	lapmod	3	4.3	4.28	4.36
int_uniform	4 blocks	500x1500	auto	3	4.34	4.25	4.37
int_uniform	4 blocks	500x1500	sap_dense	3	6.31	6.2	6.33

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	4 blocks	500x1500	ramshaw_tarjan	3	7.52	7.39	7.54
int_uniform	4 blocks	500x1500	csflow	3	9.39	9.36	9.42
int_uniform	4 blocks	500x1500	sap	3	9.54	9.39	9.68
int_uniform	4 blocks	500x1500	network_simplex	3	89.1	88.9	89.3
int_uniform	4 blocks	500x1500	csa	3	480	399	493
int_uniform	4 blocks	500x1500	auction_scaled	3	527	458	538
int_uniform	4 blocks	500x5000	hungarian	3	11.9	11.8	11.9
int_uniform	4 blocks	500x5000	jv	3	12	11.9	12
int_uniform	4 blocks	500x5000	auto	3	13.4	13.4	13.5
int_uniform	4 blocks	500x5000	lapmod	3	13.5	13.4	13.5
int_uniform	4 blocks	500x5000	sap_dense	3	19.8	19.8	19.9
int_uniform	4 blocks	500x5000	ramshaw_tarjan	3	23.7	23.7	23.8
int_uniform	4 blocks	500x5000	csflow	3	30.2	30.1	30.2
int_uniform	4 blocks	500x5000	sap	3	30.7	30.6	30.8
int_uniform	4 blocks	500x5000	network_simplex	3	507	507	521
int_uniform	4 blocks	500x5000	csa	3	4.88e+03	4.6e+03	4.93e+03
int_uniform	4 blocks	500x5000	auction_scaled	3	6.11e+03	6.07e+03	6.96e+03
int_uniform	complete	500x500	jv	3	4.45	4.05	4.49
int_uniform	complete	500x500	auto	3	4.54	4.11	4.55
int_uniform	complete	500x500	hungarian	3	8.46	8.39	8.66
int_uniform	complete	500x500	csa	3	9.81	9.7	9.94
int_uniform	complete	500x500	auction_scaled	3	9.93	9.82	10.1
int_uniform	complete	500x500	sap_dense	3	11.3	11.1	11.6
int_uniform	complete	500x500	lapmod	3	12	11.8	12.3
int_uniform	complete	500x500	ramshaw_tarjan	3	14.3	14.3	14.5
int_uniform	complete	500x500	sap	3	20.3	20.2	20.6
int_uniform	complete	500x500	csflow	3	20.4	20.3	20.7
int_uniform	complete	500x500	network_simplex	3	95.7	89.7	96.6
int_uniform	complete	500x1500	hungarian	3	4.14	4.11	4.2
int_uniform	complete	500x1500	jv	3	4.16	4.14	4.2
int_uniform	complete	500x1500	lapmod	3	5.57	5.53	5.64
int_uniform	complete	500x1500	sap_dense	3	6.06	5.98	6.09
int_uniform	complete	500x1500	ramshaw_tarjan	3	14.5	14.5	14.5
int_uniform	complete	500x1500	csflow	3	26.9	26.7	27.1
int_uniform	complete	500x1500	sap	3	27.2	27.1	27.2
int_uniform	complete	500x1500	auto	3	27.2	27.1	27.3
int_uniform	complete	500x1500	network_simplex	3	87.3	86.4	92.9
int_uniform	complete	500x1500	csa	3	389	383	425
int_uniform	complete	500x1500	auction_scaled	3	534	514	596
int_uniform	complete	500x5000	hungarian	3	11.9	11.8	11.9
int_uniform	complete	500x5000	jv	3	12.1	12.1	12.2
int_uniform	complete	500x5000	lapmod	3	17	17	17.1
int_uniform	complete	500x5000	sap_dense	3	18.4	18.4	18.5
int_uniform	complete	500x5000	ramshaw_tarjan	3	46.6	46.6	47
int_uniform	complete	500x5000	sap	3	99.6	99.5	100
int_uniform	complete	500x5000	csflow	3	101	100	101
int_uniform	complete	500x5000	auto	3	101	101	101
int_uniform	complete	500x5000	network_simplex	3	541	538	628
int_uniform	complete	500x5000	csa	3	4.71e+03	4.7e+03	4.84e+03
int_uniform	complete	500x5000	auction_scaled	3	4.76e+03	4.48e+03	5.26e+03
int_uniform	25%	500x500	auction_scaled	3	4.43	4.43	4.54

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	25%	500x500	csa	3	4.71	4.68	4.82
int_uniform	25%	500x500	jv	3	6.75	6.74	7.75
int_uniform	25%	500x500	hungarian	3	6.77	6.75	7.75
int_uniform	25%	500x500	lapmod	3	7.68	7.6	9.04
int_uniform	25%	500x500	auto	3	7.69	7.64	9.02
int_uniform	25%	500x500	csflow	3	8.95	8.9	9.57
int_uniform	25%	500x500	sap	3	9.02	8.9	9.65
int_uniform	25%	500x500	sap_dense	3	13.1	13	15.5
int_uniform	25%	500x500	ramshaw_tarjan	3	13.6	13.6	15.9
int_uniform	25%	500x500	network_simplex	3	102	97.1	103
int_uniform	25%	500x1500	hungarian	3	6.22	6.21	6.29
int_uniform	25%	500x1500	jv	3	6.32	6.25	6.34
int_uniform	25%	500x1500	lapmod	3	7.13	7.1	7.15
int_uniform	25%	500x1500	auto	3	7.17	7.17	7.19
int_uniform	25%	500x1500	sap_dense	3	10.2	10.1	10.2
int_uniform	25%	500x1500	ramshaw_tarjan	3	11.4	11.4	11.5
int_uniform	25%	500x1500	csflow	3	12.6	12.5	12.7
int_uniform	25%	500x1500	sap	3	12.7	12.7	12.8
int_uniform	25%	500x1500	network_simplex	3	102	101	129
int_uniform	25%	500x1500	auction_scaled	3	365	354	455
int_uniform	25%	500x1500	csa	3	463	387	495
int_uniform	25%	500x5000	hungarian	3	19.5	19.5	19.6
int_uniform	25%	500x5000	jv	3	19.7	19.7	19.7
int_uniform	25%	500x5000	lapmod	3	23.1	23	23.2
int_uniform	25%	500x5000	auto	3	23.2	23.1	23.2
int_uniform	25%	500x5000	sap_dense	3	32.7	32.7	32.8
int_uniform	25%	500x5000	ramshaw_tarjan	3	35.9	35.9	36
int_uniform	25%	500x5000	csflow	3	41.2	41.1	41.2
int_uniform	25%	500x5000	sap	3	41.7	41.6	41.8
int_uniform	25%	500x5000	network_simplex	3	587	578	602
int_uniform	25%	500x5000	auction_scaled	3	5.5e+03	5.42e+03	6.25e+03
int_uniform	25%	500x5000	csa	3	6.17e+03	5.56e+03	6.31e+03
int_uniform	60%	500x500	auction_scaled	3	8.17	8.16	8.19
int_uniform	60%	500x500	csa	3	8.39	8.38	8.46
int_uniform	60%	500x500	jv	3	8.8	8.75	8.93
int_uniform	60%	500x500	hungarian	3	8.8	8.75	8.93
int_uniform	60%	500x500	auto	3	8.82	8.77	8.93
int_uniform	60%	500x500	lapmod	3	12.6	12.1	12.7
int_uniform	60%	500x500	sap	3	16.2	16.1	16.4
int_uniform	60%	500x500	csflow	3	16.2	16.1	16.4
int_uniform	60%	500x500	sap_dense	3	19.2	18.9	19.4
int_uniform	60%	500x500	ramshaw_tarjan	3	21.6	21.4	22.1
int_uniform	60%	500x500	network_simplex	3	85.8	85.1	94.8
int_uniform	60%	500x1500	hungarian	3	7.96	7.91	8.16
int_uniform	60%	500x1500	jv	3	8.06	8.05	8.12
int_uniform	60%	500x1500	lapmod	3	9.68	9.58	9.77
int_uniform	60%	500x1500	sap_dense	3	13.2	13.1	13.4
int_uniform	60%	500x1500	ramshaw_tarjan	3	17.8	17.6	18
int_uniform	60%	500x1500	csflow	3	23.2	23.1	23.5
int_uniform	60%	500x1500	auto	3	23.5	23.5	23.7
int_uniform	60%	500x1500	sap	3	23.6	23.4	23.8

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	60%	500x1500	network_simplex	3	130	124	145
int_uniform	60%	500x1500	csa	3	369	358	385
int_uniform	60%	500x1500	auction_scaled	3	445	416	452
int_uniform	60%	500x5000	hungarian	3	24.7	24.7	24.8
int_uniform	60%	500x5000	jv	3	25.2	25.1	25.3
int_uniform	60%	500x5000	lapmod	3	30.9	30.8	31
int_uniform	60%	500x5000	sap_dense	3	41.4	41.4	41.4
int_uniform	60%	500x5000	ramshaw_tarjan	3	55.5	55.4	55.6
int_uniform	60%	500x5000	csflow	3	74.8	74.7	75.4
int_uniform	60%	500x5000	sap	3	75	75	75.2
int_uniform	60%	500x5000	auto	3	75.9	75.8	76
int_uniform	60%	500x5000	network_simplex	3	623	621	679
int_uniform	60%	500x5000	csa	3	4.51e+03	4.46e+03	4.7e+03
int_uniform	60%	500x5000	auction_scaled	3	5.39e+03	5.17e+03	5.61e+03
metric_clustered	4 blocks	500x500	auction_scaled	3	2.73	2.7	2.75
metric_clustered	4 blocks	500x500	csa	3	3.35	3.35	3.4
metric_clustered	4 blocks	500x500	jv	3	5.05	4.75	5.06
metric_clustered	4 blocks	500x500	hungarian	3	5.05	4.77	5.05
metric_clustered	4 blocks	500x500	lapmod	3	5.87	5.54	5.9
metric_clustered	4 blocks	500x500	auto	3	5.91	5.56	5.92
metric_clustered	4 blocks	500x500	csflow	3	6.06	5.93	6.16
metric_clustered	4 blocks	500x500	sap	3	6.1	6.01	6.26
metric_clustered	4 blocks	500x500	ramshaw_tarjan	3	6.41	6.17	6.5
metric_clustered	4 blocks	500x500	sap_dense	3	7.43	7.05	7.45
metric_clustered	4 blocks	500x500	network_simplex	3	19.3	19	20.5
metric_clustered	4 blocks	500x1500	hungarian	3	3.84	3.81	3.87
metric_clustered	4 blocks	500x1500	jv	3	3.85	3.83	3.87
metric_clustered	4 blocks	500x1500	lapmod	3	4.44	4.41	4.52
metric_clustered	4 blocks	500x1500	auto	3	4.47	4.45	4.55
metric_clustered	4 blocks	500x1500	sap_dense	3	6.34	6.32	6.35
metric_clustered	4 blocks	500x1500	ramshaw_tarjan	3	7.28	7.17	7.36
metric_clustered	4 blocks	500x1500	sap	3	8.95	8.91	9.01
metric_clustered	4 blocks	500x1500	csflow	3	8.98	8.95	9.12
metric_clustered	4 blocks	500x1500	network_simplex	3	83.4	81.5	88.7
metric_clustered	4 blocks	500x1500	auction_scaled	3	354	350	370
metric_clustered	4 blocks	500x1500	csa	3	654	594	661
metric_clustered	4 blocks	500x5000	jv	3	12.1	12	15.5
metric_clustered	4 blocks	500x5000	lapmod	3	13.6	13.5	18
metric_clustered	4 blocks	500x5000	auto	3	13.9	13.7	18.1
metric_clustered	4 blocks	500x5000	hungarian	3	18.1	15.8	18.5
metric_clustered	4 blocks	500x5000	sap_dense	3	28.2	25.8	29.2
metric_clustered	4 blocks	500x5000	csflow	3	29.3	29.2	29.5
metric_clustered	4 blocks	500x5000	sap	3	29.8	29.6	29.9
metric_clustered	4 blocks	500x5000	ramshaw_tarjan	3	30.1	26.7	30.9
metric_clustered	4 blocks	500x5000	network_simplex	3	598	596	600
metric_clustered	4 blocks	500x5000	auction_scaled	3	4.61e+03	4.49e+03	4.64e+03
metric_clustered	4 blocks	500x5000	csa	3	1.34e+04	1.21e+04	1.4e+04
metric_clustered	complete	500x500	jv	3	6.25	6.05	7.08
metric_clustered	complete	500x500	auto	3	6.25	6.04	7.08
metric_clustered	complete	500x500	auction_scaled	3	7.74	7.54	8.33
metric_clustered	complete	500x500	csa	3	9.53	9.36	10.3

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_clustered	complete	500x500	hungarian	3	9.88	9.47	11.4
metric_clustered	complete	500x500	sap_dense	3	13.2	12.7	15.4
metric_clustered	complete	500x500	lapmod	3	14.1	13.5	16.6
metric_clustered	complete	500x500	ramshaw_tarjan	3	17.6	16.8	19.2
metric_clustered	complete	500x500	sap	3	23.6	22.8	25.7
metric_clustered	complete	500x500	csflow	3	23.7	22.8	25.8
metric_clustered	complete	500x500	network_simplex	3	35	34	36.5
metric_clustered	complete	500x1500	jv	3	3.91	3.85	3.92
metric_clustered	complete	500x1500	hungarian	3	3.91	3.85	3.91
metric_clustered	complete	500x1500	lapmod	3	5.46	5.37	5.47
metric_clustered	complete	500x1500	sap_dense	3	6.07	6	6.07
metric_clustered	complete	500x1500	ramshaw_tarjan	3	13.3	13.2	13.3
metric_clustered	complete	500x1500	csflow	3	24.9	24.8	25.1
metric_clustered	complete	500x1500	auto	3	25.4	25.2	25.5
metric_clustered	complete	500x1500	sap	3	25.5	25.2	25.8
metric_clustered	complete	500x1500	network_simplex	3	98	97.7	103
metric_clustered	complete	500x1500	auction_scaled	3	350	341	379
metric_clustered	complete	500x1500	csa	3	512	498	611
metric_clustered	complete	500x5000	jv	3	11.1	11	14.8
metric_clustered	complete	500x5000	hungarian	3	11.1	11	11.2
metric_clustered	complete	500x5000	lapmod	3	16.7	16.6	16.7
metric_clustered	complete	500x5000	sap_dense	3	21.4	21.3	21.7
metric_clustered	complete	500x5000	ramshaw_tarjan	3	45.8	45.7	49.7
metric_clustered	complete	500x5000	csflow	3	97.9	97.1	98.3
metric_clustered	complete	500x5000	sap	3	98.9	98	104
metric_clustered	complete	500x5000	auto	3	107	107	109
metric_clustered	complete	500x5000	network_simplex	3	614	603	648
metric_clustered	complete	500x5000	auction_scaled	3	4.1e+03	4.09e+03	4.54e+03
metric_clustered	complete	500x5000	csa	3	1.05e+04	1.04e+04	1.21e+04
metric_clustered	25%	500x500	auction_scaled	3	4.53	4.52	4.55
metric_clustered	25%	500x500	csa	3	5.65	5.63	5.77
metric_clustered	25%	500x500	hungarian	3	13.6	12.9	14.1
metric_clustered	25%	500x500	jv	3	13.6	12.9	14.1
metric_clustered	25%	500x500	sap	3	14.4	13.9	14.7
metric_clustered	25%	500x500	csflow	3	14.4	13.9	14.7
metric_clustered	25%	500x500	auto	3	16.1	15.3	16.8
metric_clustered	25%	500x500	lapmod	3	16.2	15.3	16.8
metric_clustered	25%	500x500	sap_dense	3	28.1	26.7	29.5
metric_clustered	25%	500x500	ramshaw_tarjan	3	28.8	27.4	30.2
metric_clustered	25%	500x500	network_simplex	3	52.6	51.5	52.8
metric_clustered	25%	500x1500	jv	3	6.6	6.59	6.66
metric_clustered	25%	500x1500	hungarian	3	6.67	6.66	6.7
metric_clustered	25%	500x1500	lapmod	3	7.36	7.33	7.39
metric_clustered	25%	500x1500	auto	3	7.39	7.38	7.44
metric_clustered	25%	500x1500	sap_dense	3	10.6	10.6	10.6
metric_clustered	25%	500x1500	ramshaw_tarjan	3	12	12	12.1
metric_clustered	25%	500x1500	csflow	3	13.2	13.2	13.3
metric_clustered	25%	500x1500	sap	3	13.4	13.3	13.5
metric_clustered	25%	500x1500	network_simplex	3	112	108	112
metric_clustered	25%	500x1500	auction_scaled	3	340	328	358
metric_clustered	25%	500x1500	csa	3	587	534	599

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_clustered	25%	500x5000	auto	3	23.2	23.2	23.3
metric_clustered	25%	500x5000	jv	3	26.6	23.2	26.7
metric_clustered	25%	500x5000	hungarian	3	28.8	24.3	29.5
metric_clustered	25%	500x5000	lapmod	3	32	31.9	32
metric_clustered	25%	500x5000	ramshaw_tarjan	3	36.8	36.5	42.7
metric_clustered	25%	500x5000	csflow	3	41.5	41.2	41.6
metric_clustered	25%	500x5000	sap_dense	3	42.3	40.9	43.2
metric_clustered	25%	500x5000	sap	3	45.2	44	45.8
metric_clustered	25%	500x5000	network_simplex	3	664	657	672
metric_clustered	25%	500x5000	auction_scaled	3	4.14e+03	4.07e+03	4.96e+03
metric_clustered	25%	500x5000	csa	3	1.02e+04	1.02e+04	1.1e+04
metric_clustered	60%	500x500	auction_scaled	3	7.61	7.58	7.65
metric_clustered	60%	500x500	csa	3	9.6	9.49	9.64
metric_clustered	60%	500x500	hungarian	3	13.6	12.1	14
metric_clustered	60%	500x500	jv	3	13.6	12.1	14
metric_clustered	60%	500x500	auto	3	13.7	12.1	14.1
metric_clustered	60%	500x500	lapmod	3	20.8	17.8	21.1
metric_clustered	60%	500x500	sap	3	22.1	20.4	22.6
metric_clustered	60%	500x500	csflow	3	22.2	20.4	22.6
metric_clustered	60%	500x500	sap_dense	3	30.5	26.4	31.3
metric_clustered	60%	500x500	ramshaw_tarjan	3	34.4	30.1	35.5
metric_clustered	60%	500x500	network_simplex	3	37.4	35.9	37.6
metric_clustered	60%	500x1500	jv	3	8.4	8.35	8.46
metric_clustered	60%	500x1500	hungarian	3	8.55	8.4	8.56
metric_clustered	60%	500x1500	lapmod	3	10.1	10	10.1
metric_clustered	60%	500x1500	sap_dense	3	13.6	13.6	13.7
metric_clustered	60%	500x1500	ramshaw_tarjan	3	18.6	18.5	18.8
metric_clustered	60%	500x1500	sap	3	24.3	24.2	24.5
metric_clustered	60%	500x1500	csflow	3	24.3	24.2	24.4
metric_clustered	60%	500x1500	auto	3	24.3	24.3	24.5
metric_clustered	60%	500x1500	network_simplex	3	118	116	118
metric_clustered	60%	500x1500	auction_scaled	3	349	345	377
metric_clustered	60%	500x1500	csa	3	631	563	644
metric_clustered	60%	500x5000	jv	3	25.1	25.1	25.2
metric_clustered	60%	500x5000	hungarian	3	26.9	26	28.4
metric_clustered	60%	500x5000	lapmod	3	30.7	30.7	35.2
metric_clustered	60%	500x5000	sap_dense	3	50.6	48	51.1
metric_clustered	60%	500x5000	ramshaw_tarjan	3	55.5	55.5	58.8
metric_clustered	60%	500x5000	auto	3	75.7	75.6	82.2
metric_clustered	60%	500x5000	csflow	3	76.4	76.1	76.6
metric_clustered	60%	500x5000	sap	3	78.9	77.6	83.8
metric_clustered	60%	500x5000	network_simplex	3	665	664	665
metric_clustered	60%	500x5000	auction_scaled	3	4.25e+03	4.12e+03	4.3e+03
metric_clustered	60%	500x5000	csa	3	1.04e+04	1.02e+04	1.04e+04
metric_uniform	4 blocks	500x500	auction_scaled	3	2.48	2.44	2.48
metric_uniform	4 blocks	500x500	csa	3	3.12	3.12	3.17
metric_uniform	4 blocks	500x500	jv	3	4.15	4.14	4.3
metric_uniform	4 blocks	500x500	hungarian	3	4.19	4.16	4.32
metric_uniform	4 blocks	500x500	lapmod	3	4.84	4.83	5
metric_uniform	4 blocks	500x500	auto	3	4.85	4.83	5.01
metric_uniform	4 blocks	500x500	sap	3	5.48	5.46	5.58

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	4 blocks	500x500	csflow	3	5.51	5.48	5.58
metric_uniform	4 blocks	500x500	ramshaw_tarjan	3	5.66	5.58	5.74
metric_uniform	4 blocks	500x500	sap_dense	3	6.12	6.12	6.37
metric_uniform	4 blocks	500x500	network_simplex	3	25.3	24.2	25.8
metric_uniform	4 blocks	500x1500	jv	3	3.64	3.62	3.66
metric_uniform	4 blocks	500x1500	hungarian	3	3.66	3.63	3.67
metric_uniform	4 blocks	500x1500	auto	3	4.28	4.25	4.29
metric_uniform	4 blocks	500x1500	lapmod	3	4.29	4.26	4.32
metric_uniform	4 blocks	500x1500	sap_dense	3	6.03	6.02	6.05
metric_uniform	4 blocks	500x1500	ramshaw_tarjan	3	7.41	7.4	7.48
metric_uniform	4 blocks	500x1500	csflow	3	9.31	9.3	9.35
metric_uniform	4 blocks	500x1500	sap	3	9.42	9.38	9.45
metric_uniform	4 blocks	500x1500	network_simplex	3	89.1	85.6	90.1
metric_uniform	4 blocks	500x1500	auction_scaled	3	322	304	333
metric_uniform	4 blocks	500x1500	csa	3	595	510	600
metric_uniform	4 blocks	500x5000	auto	3	13.5	13.4	17.9
metric_uniform	4 blocks	500x5000	hungarian	3	18.2	18.1	18.5
metric_uniform	4 blocks	500x5000	jv	3	18.7	15.3	18.9
metric_uniform	4 blocks	500x5000	lapmod	3	20.2	18.3	21.2
metric_uniform	4 blocks	500x5000	ramshaw_tarjan	3	23.9	23.8	28.1
metric_uniform	4 blocks	500x5000	csflow	3	30.2	30.2	30.5
metric_uniform	4 blocks	500x5000	sap	3	31	30.8	34.9
metric_uniform	4 blocks	500x5000	sap_dense	3	31.6	27.6	34.4
metric_uniform	4 blocks	500x5000	network_simplex	3	509	507	516
metric_uniform	4 blocks	500x5000	auction_scaled	3	4.27e+03	4.15e+03	4.74e+03
metric_uniform	4 blocks	500x5000	csa	3	1.17e+04	1.11e+04	1.17e+04
metric_uniform	complete	500x500	auto	3	5.7	5.01	5.95
metric_uniform	complete	500x500	jv	3	5.7	5.01	5.95
metric_uniform	complete	500x500	auction_scaled	3	6.98	6.97	7.07
metric_uniform	complete	500x500	csa	3	9.11	8.98	9.24
metric_uniform	complete	500x500	hungarian	3	9.43	8.49	9.85
metric_uniform	complete	500x500	sap_dense	3	12.5	11.3	13.1
metric_uniform	complete	500x500	lapmod	3	13.3	11.8	14
metric_uniform	complete	500x500	ramshaw_tarjan	3	15.9	14.9	16.4
metric_uniform	complete	500x500	csflow	3	22.4	21	23
metric_uniform	complete	500x500	sap	3	22.4	21.1	23
metric_uniform	complete	500x500	network_simplex	3	67.5	65.1	79.5
metric_uniform	complete	500x1500	jv	3	3.7	3.68	3.84
metric_uniform	complete	500x1500	hungarian	3	3.72	3.69	3.83
metric_uniform	complete	500x1500	lapmod	3	5.18	5.16	5.32
metric_uniform	complete	500x1500	sap_dense	3	6.05	5.95	6.05
metric_uniform	complete	500x1500	ramshaw_tarjan	3	14.1	14.1	14.4
metric_uniform	complete	500x1500	csflow	3	26.7	26.5	27
metric_uniform	complete	500x1500	sap	3	26.7	26.7	27.1
metric_uniform	complete	500x1500	auto	3	26.8	26.7	27.3
metric_uniform	complete	500x1500	network_simplex	3	112	102	177
metric_uniform	complete	500x1500	auction_scaled	3	326	304	350
metric_uniform	complete	500x1500	csa	3	416	416	544
metric_uniform	complete	500x5000	jv	3	11	10.9	11.1
metric_uniform	complete	500x5000	hungarian	3	11.3	11.2	14.7
metric_uniform	complete	500x5000	lapmod	3	16.4	16.3	16.5

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	complete	500x5000	sap_dense	3	21.2	21.2	21.4
metric_uniform	complete	500x5000	ramshaw_tarjan	3	46.1	45.9	50.4
metric_uniform	complete	500x5000	sap	3	100	99	104
metric_uniform	complete	500x5000	csflow	3	100	99.4	100
metric_uniform	complete	500x5000	auto	3	110	109	111
metric_uniform	complete	500x5000	network_simplex	3	580	552	624
metric_uniform	complete	500x5000	auction_scaled	3	3.97e+03	3.96e+03	4.1e+03
metric_uniform	complete	500x5000	csa	3	1.18e+04	1.15e+04	1.19e+04
metric_uniform	25%	500x500	auction_scaled	3	4.03	4	4.08
metric_uniform	25%	500x500	csa	3	5.15	5.13	5.18
metric_uniform	25%	500x500	hungarian	3	10.6	9.65	10.9
metric_uniform	25%	500x500	jv	3	10.6	9.66	10.9
metric_uniform	25%	500x500	csflow	3	11.9	11.2	12
metric_uniform	25%	500x500	sap	3	12	11.2	12
metric_uniform	25%	500x500	lapmod	3	12.5	11.2	13
metric_uniform	25%	500x500	auto	3	12.5	11.2	13
metric_uniform	25%	500x500	sap_dense	3	21.7	19.5	22.5
metric_uniform	25%	500x500	ramshaw_tarjan	3	22.2	19.9	22.7
metric_uniform	25%	500x500	network_simplex	3	85.4	84.6	90.3
metric_uniform	25%	500x1500	jv	3	6.3	6.29	6.32
metric_uniform	25%	500x1500	hungarian	3	6.38	6.35	6.42
metric_uniform	25%	500x1500	lapmod	3	7.05	7.01	7.11
metric_uniform	25%	500x1500	auto	3	7.18	7.17	7.2
metric_uniform	25%	500x1500	sap_dense	3	10.1	10.1	10.2
metric_uniform	25%	500x1500	ramshaw_tarjan	3	11.5	11.5	11.5
metric_uniform	25%	500x1500	csflow	3	12.7	12.6	12.7
metric_uniform	25%	500x1500	sap	3	12.8	12.7	12.8
metric_uniform	25%	500x1500	network_simplex	3	135	125	165
metric_uniform	25%	500x1500	auction_scaled	3	296	296	301
metric_uniform	25%	500x1500	csa	3	671	607	695
metric_uniform	25%	500x5000	jv	3	20.1	20	23.4
metric_uniform	25%	500x5000	auto	3	23.3	23.3	26.7
metric_uniform	25%	500x5000	hungarian	3	23.6	22.2	26.5
metric_uniform	25%	500x5000	lapmod	3	31.9	31	31.9
metric_uniform	25%	500x5000	csflow	3	41.7	41.5	42.4
metric_uniform	25%	500x5000	ramshaw_tarjan	3	43.5	39.9	44.3
metric_uniform	25%	500x5000	sap	3	43.6	43	48.8
metric_uniform	25%	500x5000	sap_dense	3	50.5	49.2	50.6
metric_uniform	25%	500x5000	network_simplex	3	648	638	688
metric_uniform	25%	500x5000	auction_scaled	3	4e+03	3.89e+03	4.19e+03
metric_uniform	25%	500x5000	csa	3	1.2e+04	1.14e+04	1.21e+04
metric_uniform	60%	500x500	auction_scaled	3	6.61	6.59	6.78
metric_uniform	60%	500x500	csa	3	8.72	8.67	8.94
metric_uniform	60%	500x500	auto	3	10	9.38	10.4
metric_uniform	60%	500x500	hungarian	3	10	9.37	10.4
metric_uniform	60%	500x500	jv	3	10	9.4	10.4
metric_uniform	60%	500x500	lapmod	3	14.4	13.2	15
metric_uniform	60%	500x500	csflow	3	17.5	16.8	17.8
metric_uniform	60%	500x500	sap	3	17.6	16.8	17.9
metric_uniform	60%	500x500	sap_dense	3	21.7	20.1	22.7
metric_uniform	60%	500x500	ramshaw_tarjan	3	24.5	22.9	25.6

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	60%	500x500	network_simplex	3	77.8	76.7	78.8
metric_uniform	60%	500x1500	jv	3	7.88	7.84	7.95
metric_uniform	60%	500x1500	hungarian	3	7.93	7.84	7.98
metric_uniform	60%	500x1500	lapmod	3	9.55	9.53	9.59
metric_uniform	60%	500x1500	sap_dense	3	13	12.9	13
metric_uniform	60%	500x1500	ramshaw_tarjan	3	17.6	17.4	17.7
metric_uniform	60%	500x1500	csflow	3	23.1	23	23.4
metric_uniform	60%	500x1500	sap	3	23.2	23.2	23.5
metric_uniform	60%	500x1500	auto	3	23.4	23.2	23.5
metric_uniform	60%	500x1500	network_simplex	3	215	179	247
metric_uniform	60%	500x1500	auction_scaled	3	279	273	301
metric_uniform	60%	500x1500	csa	3	528	511	540
metric_uniform	60%	500x5000	hungarian	3	24.9	24.8	25.8
metric_uniform	60%	500x5000	lapmod	3	30.5	30.5	35.3
metric_uniform	60%	500x5000	jv	3	31.8	28.5	31.8
metric_uniform	60%	500x5000	sap_dense	3	46.8	45.8	49.2
metric_uniform	60%	500x5000	ramshaw_tarjan	3	62.5	62.2	63.1
metric_uniform	60%	500x5000	csflow	3	75.3	75.1	75.7
metric_uniform	60%	500x5000	auto	3	76.9	76.2	77.9
metric_uniform	60%	500x5000	sap	3	88.4	82.1	89.5
metric_uniform	60%	500x5000	network_simplex	3	629	616	664
metric_uniform	60%	500x5000	auction_scaled	3	3.53e+03	3.51e+03	3.79e+03
metric_uniform	60%	500x5000	csa	3	1.04e+04	1.02e+04	1.08e+04
tied	4 blocks	500x500	auction_scaled	3	4.05	4.04	4.12
tied	4 blocks	500x500	csa	3	4.23	4.21	4.29
tied	4 blocks	500x500	network_simplex	3	5.24	5.2	5.42
tied	4 blocks	500x500	sap	3	13.2	13.1	13.2
tied	4 blocks	500x500	csflow	3	13.2	13.2	13.2
tied	4 blocks	500x500	ramshaw_tarjan	3	15.2	15.1	15.2
tied	4 blocks	500x500	jv	3	17.4	17.4	17.4
tied	4 blocks	500x500	hungarian	3	17.5	17.4	17.5
tied	4 blocks	500x500	auto	3	18.7	18.7	18.7
tied	4 blocks	500x500	lapmod	3	18.7	18.6	18.8
tied	4 blocks	500x500	sap_dense	3	25.1	25.1	25.1
tied	4 blocks	500x1500	network_simplex	3	30	29.3	30.1
tied	4 blocks	500x1500	csflow	3	30.5	30.5	30.7
tied	4 blocks	500x1500	sap	3	30.6	30.6	30.6
tied	4 blocks	500x1500	ramshaw_tarjan	3	42.8	42.7	42.9
tied	4 blocks	500x1500	hungarian	3	52.8	52.8	52.8
tied	4 blocks	500x1500	jv	3	52.8	52.8	52.9
tied	4 blocks	500x1500	auto	3	57.2	57.1	57.3
tied	4 blocks	500x1500	lapmod	3	57.2	57.2	57.3
tied	4 blocks	500x1500	sap_dense	3	75.3	75.3	75.7
tied	4 blocks	500x1500	csa	3	103	103	106
tied	4 blocks	500x1500	auction_scaled	3	168	166	170
tied	4 blocks	500x5000	sap	3	91.2	91	91.5
tied	4 blocks	500x5000	csflow	3	91.5	91.3	91.7
tied	4 blocks	500x5000	ramshaw_tarjan	3	138	138	139
tied	4 blocks	500x5000	jv	3	177	176	177
tied	4 blocks	500x5000	hungarian	3	177	176	177
tied	4 blocks	500x5000	auto	3	191	190	191

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	4 blocks	500x5000	lapmod	3	191	190	191
tied	4 blocks	500x5000	network_simplex	3	195	192	198
tied	4 blocks	500x5000	sap_dense	3	254	252	255
tied	4 blocks	500x5000	csa	3	1.69e+03	1.68e+03	1.69e+03
tied	4 blocks	500x5000	auction_scaled	3	2.67e+03	2.6e+03	2.74e+03
tied	complete	500x500	network_simplex	3	5.56	5.11	5.62
tied	complete	500x500	csa	3	13.5	13.3	14.2
tied	complete	500x500	auction_scaled	3	13.7	13.5	14.4
tied	complete	500x500	jv	3	56.1	55.9	56.1
tied	complete	500x500	auto	3	57.1	56.6	57.5
tied	complete	500x500	hungarian	3	61.4	61	61.5
tied	complete	500x500	ramshaw_tarjan	3	69	68.9	69
tied	complete	500x500	sap_dense	3	90.1	89.9	90.2
tied	complete	500x500	lapmod	3	96.5	96.3	96.9
tied	complete	500x500	sap	3	113	113	114
tied	complete	500x500	csflow	3	114	114	115
tied	complete	500x1500	network_simplex	3	20.4	20.2	20.8
tied	complete	500x1500	csa	3	92.9	92.3	105
tied	complete	500x1500	auction_scaled	3	149	146	151
tied	complete	500x1500	hungarian	3	199	199	199
tied	complete	500x1500	jv	3	199	199	199
tied	complete	500x1500	ramshaw_tarjan	3	226	226	226
tied	complete	500x1500	sap	3	273	273	273
tied	complete	500x1500	auto	3	273	273	273
tied	complete	500x1500	csflow	3	276	276	276
tied	complete	500x1500	sap_dense	3	287	287	287
tied	complete	500x1500	lapmod	3	297	296	297
tied	complete	500x5000	network_simplex	3	119	117	121
tied	complete	500x5000	jv	3	681	681	682
tied	complete	500x5000	hungarian	3	681	680	683
tied	complete	500x5000	ramshaw_tarjan	3	810	807	822
tied	complete	500x5000	auto	3	864	863	864
tied	complete	500x5000	csflow	3	873	873	873
tied	complete	500x5000	sap	3	921	921	921
tied	complete	500x5000	sap_dense	3	979	978	979
tied	complete	500x5000	lapmod	3	1.01e+03	1.01e+03	1.01e+03
tied	complete	500x5000	csa	3	1.41e+03	1.34e+03	1.42e+03
tied	complete	500x5000	auction_scaled	3	1.82e+03	1.81e+03	1.88e+03
tied	25%	500x500	auction_scaled	3	5.87	5.86	5.91
tied	25%	500x500	csa	3	6.2	6.18	6.24
tied	25%	500x500	network_simplex	3	8.6	8.4	8.62
tied	25%	500x500	sap	3	52	51.8	52
tied	25%	500x500	csflow	3	52.1	52	52.2
tied	25%	500x500	hungarian	3	62	61.5	62.1
tied	25%	500x500	jv	3	62	61.6	62.1
tied	25%	500x500	auto	3	73	72.9	73.1
tied	25%	500x500	lapmod	3	73	72.9	73.2
tied	25%	500x500	ramshaw_tarjan	3	108	108	108
tied	25%	500x500	sap_dense	3	129	129	129
tied	25%	500x1500	network_simplex	3	35.3	34.5	35.4
tied	25%	500x1500	csa	3	98.3	97.9	100

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	25%	500x1500	sap	3	115	115	115
tied	25%	500x1500	csflow	3	115	115	115
tied	25%	500x1500	auction_scaled	3	139	129	141
tied	25%	500x1500	hungarian	3	198	198	198
tied	25%	500x1500	jv	3	198	198	198
tied	25%	500x1500	lapmod	3	222	222	222
tied	25%	500x1500	auto	3	222	222	222
tied	25%	500x1500	ramshaw_tarjan	3	395	395	396
tied	25%	500x1500	sap_dense	3	476	476	477
tied	25%	500x5000	network_simplex	3	218	215	219
tied	25%	500x5000	sap	3	344	343	345
tied	25%	500x5000	csflow	3	349	347	350
tied	25%	500x5000	jv	3	684	683	685
tied	25%	500x5000	hungarian	3	688	687	689
tied	25%	500x5000	auto	3	745	744	746
tied	25%	500x5000	lapmod	3	745	744	747
tied	25%	500x5000	csa	3	1.35e+03	1.27e+03	1.36e+03
tied	25%	500x5000	ramshaw_tarjan	3	1.39e+03	1.39e+03	1.4e+03
tied	25%	500x5000	sap_dense	3	1.69e+03	1.69e+03	1.69e+03
tied	25%	500x5000	auction_scaled	3	1.74e+03	1.71e+03	1.81e+03
tied	60%	500x500	network_simplex	3	7.55	7.38	7.67
tied	60%	500x500	auction_scaled	3	11.4	10.9	11.5
tied	60%	500x500	csa	3	11.7	11.1	11.8
tied	60%	500x500	auto	3	62.5	62.4	62.7
tied	60%	500x500	jv	3	63.1	63	63.2
tied	60%	500x500	hungarian	3	63.3	63.2	63.5
tied	60%	500x500	sap	3	82.8	82.8	82.9
tied	60%	500x500	csflow	3	83.4	83.3	83.4
tied	60%	500x500	lapmod	3	84.9	84.9	85.1
tied	60%	500x500	sap_dense	3	145	145	145
tied	60%	500x500	ramshaw_tarjan	3	149	149	149
tied	60%	500x1500	network_simplex	3	35.5	34.3	36.5
tied	60%	500x1500	csa	3	105	102	107
tied	60%	500x1500	auction_scaled	3	142	137	145
tied	60%	500x1500	auto	3	195	195	195
tied	60%	500x1500	sap	3	195	195	195
tied	60%	500x1500	csflow	3	197	196	197
tied	60%	500x1500	hungarian	3	203	203	204
tied	60%	500x1500	jv	3	205	204	205
tied	60%	500x1500	lapmod	3	262	262	262
tied	60%	500x1500	ramshaw_tarjan	3	558	558	559
tied	60%	500x1500	sap_dense	3	595	595	597
tied	60%	500x5000	network_simplex	3	181	179	181
tied	60%	500x5000	auto	3	609	608	609
tied	60%	500x5000	sap	3	609	609	610
tied	60%	500x5000	csflow	3	644	629	645
tied	60%	500x5000	hungarian	3	697	697	697
tied	60%	500x5000	jv	3	700	699	701
tied	60%	500x5000	lapmod	3	884	884	886
tied	60%	500x5000	csa	3	1.38e+03	1.28e+03	1.46e+03
tied	60%	500x5000	auction_scaled	3	1.52e+03	1.51e+03	1.7e+03

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	60%	500x5000	ramshaw_tarjan	3	1.98e+03	1.98e+03	1.99e+03
tied	60%	500x5000	sap_dense	3	2.14e+03	2.14e+03	2.14e+03
binary	complete	1000x10000	auto	2	168	167	169
binary	complete	1000x10000	jv	2	5.31e+03	5.3e+03	5.32e+03
binary	complete	1000x10000	ramshaw_tarjan	2	5.86e+03	5.86e+03	5.86e+03
binary	complete	1000x10000	sap	2	5.96e+03	5.96e+03	5.96e+03
binary	complete	1000x10000	lapmod	2	7.95e+03	7.95e+03	7.95e+03
binary	complete	1500x1500	auto	2	36.8	36.2	37.4
binary	complete	1500x1500	jv	2	1.49e+03	1.49e+03	1.49e+03
binary	complete	1500x1500	ramshaw_tarjan	2	1.51e+03	1.51e+03	1.51e+03
binary	complete	1500x1500	lapmod	2	2.52e+03	2.51e+03	2.52e+03
binary	complete	1500x1500	sap	2	2.52e+03	2.51e+03	2.52e+03
binary	complete	1500x4500	auto	2	116	115	116
binary	complete	1500x4500	jv	2	5.11e+03	5.11e+03	5.12e+03
binary	complete	1500x4500	ramshaw_tarjan	2	5.45e+03	5.44e+03	5.45e+03
binary	complete	1500x4500	sap	2	6.39e+03	6.35e+03	6.43e+03
binary	complete	1500x4500	lapmod	2	7.88e+03	7.87e+03	7.88e+03
binary	25%	1000x10000	auto	2	136	136	137
binary	25%	1000x10000	sap	2	2.05e+03	2.05e+03	2.05e+03
binary	25%	1000x10000	jv	2	5.37e+03	5.37e+03	5.38e+03
binary	25%	1000x10000	lapmod	2	5.95e+03	5.95e+03	5.95e+03
binary	25%	1000x10000	ramshaw_tarjan	2	1.04e+04	1.04e+04	1.04e+04
binary	25%	1500x1500	auto	2	26.8	26.8	26.8
binary	25%	1500x1500	sap	2	810	810	810
binary	25%	1500x1500	jv	2	1.52e+03	1.52e+03	1.52e+03
binary	25%	1500x1500	lapmod	2	1.89e+03	1.89e+03	1.89e+03
binary	25%	1500x1500	ramshaw_tarjan	2	2.66e+03	2.66e+03	2.66e+03
binary	25%	1500x4500	auto	2	93.9	93.9	93.9
binary	25%	1500x4500	sap	2	1.98e+03	1.98e+03	1.98e+03
binary	25%	1500x4500	jv	2	5.17e+03	5.17e+03	5.18e+03
binary	25%	1500x4500	lapmod	2	5.78e+03	5.78e+03	5.78e+03
binary	25%	1500x4500	ramshaw_tarjan	2	9.96e+03	9.96e+03	9.96e+03
dbl_uniform	complete	1000x10000	jv	2	77.1	76.4	77.7
dbl_uniform	complete	1000x10000	lapmod	2	99.2	98.7	99.7
dbl_uniform	complete	1000x10000	ramshaw_tarjan	2	225	224	227
dbl_uniform	complete	1000x10000	sap	2	473	473	474
dbl_uniform	complete	1000x10000	auto	2	475	475	475
dbl_uniform	complete	1500x1500	auto	2	50.5	49.8	51.2
dbl_uniform	complete	1500x1500	jv	2	50.8	49.7	52
dbl_uniform	complete	1500x1500	ramshaw_tarjan	2	158	156	159
dbl_uniform	complete	1500x1500	lapmod	2	161	160	163
dbl_uniform	complete	1500x1500	sap	2	259	257	261
dbl_uniform	complete	1500x4500	jv	2	59	58.9	59.1
dbl_uniform	complete	1500x4500	lapmod	2	73.1	72.8	73.4
dbl_uniform	complete	1500x4500	ramshaw_tarjan	2	165	165	166
dbl_uniform	complete	1500x4500	auto	2	320	319	321
dbl_uniform	complete	1500x4500	sap	2	320	319	321
dbl_uniform	25%	1000x10000	jv	2	112	111	112
dbl_uniform	25%	1000x10000	auto	2	127	125	129
dbl_uniform	25%	1000x10000	lapmod	2	128	126	129
dbl_uniform	25%	1000x10000	ramshaw_tarjan	2	188	188	189

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	25%	1000x10000	sap	2	218	217	219
dbl_uniform	25%	1500x1500	jv	2	97.3	91.8	103
dbl_uniform	25%	1500x1500	sap	2	107	105	109
dbl_uniform	25%	1500x1500	auto	2	118	110	126
dbl_uniform	25%	1500x1500	lapmod	2	118	110	126
dbl_uniform	25%	1500x1500	ramshaw_tarjan	2	213	199	226
dbl_uniform	25%	1500x4500	auto	2	88.3	86.9	89.8
dbl_uniform	25%	1500x4500	lapmod	2	92.9	89.3	96.4
dbl_uniform	25%	1500x4500	jv	2	94.6	94.5	94.7
dbl_uniform	25%	1500x4500	ramshaw_tarjan	2	134	132	136
dbl_uniform	25%	1500x4500	sap	2	151	150	152
heavy_tailed	complete	1000x10000	jv	2	77.5	77.4	77.6
heavy_tailed	complete	1000x10000	lapmod	2	98.8	98.7	99
heavy_tailed	complete	1000x10000	ramshaw_tarjan	2	224	223	224
heavy_tailed	complete	1000x10000	auto	2	465	465	466
heavy_tailed	complete	1000x10000	sap	2	466	464	467
heavy_tailed	complete	1500x1500	auto	2	50.3	47.5	53
heavy_tailed	complete	1500x1500	jv	2	50.4	49.3	51.6
heavy_tailed	complete	1500x1500	lapmod	2	142	140	143
heavy_tailed	complete	1500x1500	ramshaw_tarjan	2	145	145	146
heavy_tailed	complete	1500x1500	sap	2	238	238	238
heavy_tailed	complete	1500x4500	jv	2	62.8	60.8	64.8
heavy_tailed	complete	1500x4500	lapmod	2	72.5	72.4	72.7
heavy_tailed	complete	1500x4500	ramshaw_tarjan	2	167	166	167
heavy_tailed	complete	1500x4500	auto	2	319	318	320
heavy_tailed	complete	1500x4500	sap	2	324	322	326
heavy_tailed	25%	1000x10000	jv	2	111	111	111
heavy_tailed	25%	1000x10000	lapmod	2	123	123	123
heavy_tailed	25%	1000x10000	auto	2	123	123	123
heavy_tailed	25%	1000x10000	ramshaw_tarjan	2	187	186	189
heavy_tailed	25%	1000x10000	sap	2	214	213	214
heavy_tailed	25%	1500x1500	jv	2	101	99	102
heavy_tailed	25%	1500x1500	sap	2	108	108	108
heavy_tailed	25%	1500x1500	auto	2	117	117	118
heavy_tailed	25%	1500x1500	lapmod	2	119	119	120
heavy_tailed	25%	1500x1500	ramshaw_tarjan	2	211	210	212
heavy_tailed	25%	1500x4500	lapmod	2	85.1	85	85.2
heavy_tailed	25%	1500x4500	auto	2	85.2	85.2	85.2
heavy_tailed	25%	1500x4500	jv	2	86.5	85.9	87.1
heavy_tailed	25%	1500x4500	ramshaw_tarjan	2	130	129	131
heavy_tailed	25%	1500x4500	sap	2	153	152	154
int_uniform	complete	1000x10000	jv	2	78.6	78.4	78.8
int_uniform	complete	1000x10000	lapmod	2	99.7	99.6	99.9
int_uniform	complete	1000x10000	ramshaw_tarjan	2	225	225	225
int_uniform	complete	1000x10000	auto	2	474	472	475
int_uniform	complete	1000x10000	sap	2	476	475	477
int_uniform	complete	1500x1500	jv	2	43.8	43.4	44.3
int_uniform	complete	1500x1500	auto	2	43.9	43.5	44.4
int_uniform	complete	1500x1500	lapmod	2	129	128	131
int_uniform	complete	1500x1500	ramshaw_tarjan	2	140	138	141
int_uniform	complete	1500x1500	sap	2	225	223	227

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	complete	1500x4500	jv	2	60.6	60.3	60.9
int_uniform	complete	1500x4500	lapmod	2	75.1	74.7	75.5
int_uniform	complete	1500x4500	ramshaw_tarjan	2	162	162	162
int_uniform	complete	1500x4500	auto	2	319	318	321
int_uniform	complete	1500x4500	sap	2	321	320	321
int_uniform	25%	1000x10000	jv	2	108	108	109
int_uniform	25%	1000x10000	lapmod	2	121	121	121
int_uniform	25%	1000x10000	auto	2	122	122	122
int_uniform	25%	1000x10000	ramshaw_tarjan	2	181	180	181
int_uniform	25%	1000x10000	sap	2	213	212	213
int_uniform	25%	1500x1500	jv	2	98.4	98.2	98.6
int_uniform	25%	1500x1500	sap	2	106	105	108
int_uniform	25%	1500x1500	lapmod	2	122	121	123
int_uniform	25%	1500x1500	auto	2	122	121	122
int_uniform	25%	1500x1500	ramshaw_tarjan	2	220	217	223
int_uniform	25%	1500x4500	jv	2	78	77.5	78.5
int_uniform	25%	1500x4500	auto	2	87.3	85.8	88.9
int_uniform	25%	1500x4500	lapmod	2	88.5	87	90
int_uniform	25%	1500x4500	ramshaw_tarjan	2	127	127	127
int_uniform	25%	1500x4500	sap	2	161	156	166
metric_clustered	complete	1000x10000	jv	2	77	76.8	77.1
metric_clustered	complete	1000x10000	lapmod	2	98.4	98.3	98.5
metric_clustered	complete	1000x10000	ramshaw_tarjan	2	221	220	221
metric_clustered	complete	1000x10000	auto	2	464	463	465
metric_clustered	complete	1000x10000	sap	2	468	465	471
metric_clustered	complete	1500x1500	auto	2	153	140	166
metric_clustered	complete	1500x1500	jv	2	153	140	166
metric_clustered	complete	1500x1500	ramshaw_tarjan	2	275	260	290
metric_clustered	complete	1500x1500	lapmod	2	305	284	325
metric_clustered	complete	1500x1500	sap	2	412	391	433
metric_clustered	complete	1500x4500	jv	2	59.2	58.7	59.7
metric_clustered	complete	1500x4500	lapmod	2	77.8	76.6	79
metric_clustered	complete	1500x4500	ramshaw_tarjan	2	165	161	168
metric_clustered	complete	1500x4500	auto	2	316	314	318
metric_clustered	complete	1500x4500	sap	2	317	316	318
metric_clustered	25%	1000x10000	jv	2	113	113	113
metric_clustered	25%	1000x10000	lapmod	2	124	124	124
metric_clustered	25%	1000x10000	auto	2	127	125	129
metric_clustered	25%	1000x10000	ramshaw_tarjan	2	189	187	190
metric_clustered	25%	1000x10000	sap	2	218	218	218
metric_clustered	25%	1500x1500	sap	2	148	144	152
metric_clustered	25%	1500x1500	jv	2	156	150	161
metric_clustered	25%	1500x1500	auto	2	181	174	188
metric_clustered	25%	1500x1500	lapmod	2	182	174	191
metric_clustered	25%	1500x1500	ramshaw_tarjan	2	335	321	350
metric_clustered	25%	1500x4500	jv	2	87.4	85.2	89.6
metric_clustered	25%	1500x4500	auto	2	90.5	89.1	92
metric_clustered	25%	1500x4500	lapmod	2	102	95.3	108
metric_clustered	25%	1500x4500	ramshaw_tarjan	2	138	136	141
metric_clustered	25%	1500x4500	sap	2	164	162	166

2.4 Do the solvers agree

Every solver in a cell solves the same instance, so their totals have to agree. A disagreement would be a defect in a solver rather than a property of the regime.

348 instances carrying 5 to 12 solvers each; the worst relative disagreement on the optimum is 696.

3 The edge-generation loop away from one cloud

The article’s implicit results come from one cloud: eight independent normal covariates with the treated group shifted, one draw per size. This grid sweeps one factor at a time around that base, several seeds deep. Where the dense path can afford the problem, the loop’s answer is compared against it unit by unit.

Table 7: The loop by cell. Rounds is the range over seeds. Graph is the median share of the complete pair count the candidate set ended up holding, in percent. Distances is the median number of distance evaluations over all rounds as a multiple of the complete pair count. Implicit and lazy are median seconds over seeds and speedup is their median ratio.

Sweep	Cloud	Dim	Units	Caliper	Seeds	Rounds	Graph %	Distances	Implicit	Lazy	Speedup
caliper	gaussian	8	10,000	boundary	5	2-2	1.151	0.31x	0.911 s	1.66 s	1.79x
caliper	gaussian	8	10,000	mid	5	2-2	1.163	0.51x	0.991 s	1.8 s	1.82x
caliper	gaussian	8	10,000	none	5	2-2	1.170	1.79x	0.862 s	1.77 s	2.06x
cloud	clustered	8	10,000	none	5	2-2	1.170	0.48x	0.325 s	0.654 s	2.00x
cloud	contested	8	10,000	none	5	9-10	73.318	6.67x	133 s	426 s	3.18x
cloud	gaussian	8	10,000	none	5	2-2	1.170	1.79x	0.863 s	1.77 s	2.04x
cloud	heavy_tailed	8	10,000	none	5	2-2	1.170	1.99x	0.894 s	0.964 s	1.08x
cloud	lattice_ties	8	10,000	none	5	2-2	1.170	1.78x	1.09 s	1.96 s	1.79x
cloud	shell	8	10,000	none	5	4-7	1.183	4.96x	2.63 s	14.8 s	5.57x
cloud	shifted	8	10,000	none	5	7-7	3.086	5.92x	4.81 s	54.3 s	11.11x
dimension	gaussian	2	10,000	none	5	2-4	1.170	0.24x	0.164 s	0.6 s	3.81x
dimension	gaussian	4	10,000	none	5	2-2	1.170	0.69x	0.335 s	1.45 s	4.37x
dimension	gaussian	8	10,000	none	5	2-2	1.170	1.79x	0.864 s	1.77 s	2.05x
dimension	gaussian	16	10,000	none	5	2-2	1.170	2.00x	2.51 s	4.96 s	1.97x
dimension	gaussian	32	10,000	none	5	2-2	1.170	2.00x	9.58 s	21.3 s	2.22x
size	gaussian	8	2,000	none	3	2-2	4.951	1.99x	0.05 s	0.066 s	1.29x
size	gaussian	8	5,000	none	3	2-2	2.160	1.92x	0.25 s	0.37 s	1.48x
size	gaussian	8	10,000	none	3	2-2	1.170	1.79x	0.863 s	1.77 s	2.05x
size	gaussian	8	20,000	none	3	2-2	0.630	1.61x	2.98 s	7.7 s	2.57x
size	gaussian	8	50,000	none	3	2-3	0.288	1.94x	21.9 s	64.4 s	2.99x

Table 8: The certificate by cell. The duality gap is the residual the loop stopped on. Equals the dense solve compares the pairing unit by unit where the dense path could be run, and reads not run where the problem was too large for it.

Sweep	Cloud	Dim	Units	Caliper	Worst duality gap	All certified	Equals the dense solve
caliper	gaussian	8	10,000	boundary	0.00e+00	TRUE	TRUE
caliper	gaussian	8	10,000	mid	0.00e+00	TRUE	TRUE
caliper	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	clustered	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	contested	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	heavy_tailed	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	lattice_ties	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	shell	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	shifted	8	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	2	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	4	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	16	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	32	10,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	2,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	5,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	20,000	none	0.00e+00	TRUE	not run
size	gaussian	8	50,000	none	3.64e-12	TRUE	not run

Table 9: Every implicit run in the grid, one row per seed. Seed width is the number of columns the first round gave each row.

Sweep	Cloud	Dim	Units	Caliper	Seed	Rounds	Seed width	Graph %	Seconds	Gap
dimension	gaussian	2	10,000	none	1	2	78	1.170	0.139	0.00e+00
dimension	gaussian	2	10,000	none	2	3	78	1.170	0.172	0.00e+00
dimension	gaussian	2	10,000	none	3	3	78	1.170	0.164	0.00e+00
dimension	gaussian	2	10,000	none	4	2	78	1.170	0.143	0.00e+00
dimension	gaussian	2	10,000	none	5	4	78	1.170	0.181	0.00e+00
dimension	gaussian	4	10,000	none	1	2	78	1.170	0.333	0.00e+00
dimension	gaussian	4	10,000	none	2	2	78	1.170	0.341	0.00e+00
dimension	gaussian	4	10,000	none	3	2	78	1.170	0.34	0.00e+00
dimension	gaussian	4	10,000	none	4	2	78	1.170	0.332	0.00e+00
dimension	gaussian	4	10,000	none	5	2	78	1.170	0.335	0.00e+00
dimension	gaussian	8	10,000	none	1	2	78	1.170	0.864	0.00e+00
dimension	gaussian	8	10,000	none	2	2	78	1.170	0.862	0.00e+00
dimension	gaussian	8	10,000	none	3	2	78	1.170	0.864	0.00e+00
dimension	gaussian	8	10,000	none	4	2	78	1.170	0.866	0.00e+00
dimension	gaussian	8	10,000	none	5	2	78	1.170	0.862	0.00e+00
dimension	gaussian	16	10,000	none	1	2	78	1.170	2.51	0.00e+00
dimension	gaussian	16	10,000	none	2	2	78	1.170	2.51	0.00e+00
dimension	gaussian	16	10,000	none	3	2	78	1.170	2.51	0.00e+00
dimension	gaussian	16	10,000	none	4	2	78	1.170	2.52	0.00e+00

Sweep	Cloud	Dim	Units	Caliper	Seed	Rounds	Seed width	Graph %	Seconds	Gap
dimension	gaussian	16	10,000	none	5	2	78	1.170	2.52	0.00e+00
dimension	gaussian	32	10,000	none	1	2	78	1.170	9.58	0.00e+00
dimension	gaussian	32	10,000	none	2	2	78	1.170	9.58	0.00e+00
dimension	gaussian	32	10,000	none	3	2	78	1.170	9.58	0.00e+00
dimension	gaussian	32	10,000	none	4	2	78	1.170	9.58	0.00e+00
dimension	gaussian	32	10,000	none	5	2	78	1.170	9.58	0.00e+00
cloud	gaussian	8	10,000	none	1	2	78	1.170	0.863	0.00e+00
cloud	gaussian	8	10,000	none	2	2	78	1.170	0.864	0.00e+00
cloud	gaussian	8	10,000	none	3	2	78	1.170	0.863	0.00e+00
cloud	gaussian	8	10,000	none	4	2	78	1.170	0.867	0.00e+00
cloud	gaussian	8	10,000	none	5	2	78	1.170	0.861	0.00e+00
cloud	clustered	8	10,000	none	1	2	78	1.170	0.314	0.00e+00
cloud	clustered	8	10,000	none	2	2	78	1.170	0.325	0.00e+00
cloud	clustered	8	10,000	none	3	2	78	1.170	0.333	0.00e+00
cloud	clustered	8	10,000	none	4	2	78	1.170	0.36	0.00e+00
cloud	clustered	8	10,000	none	5	2	78	1.170	0.316	0.00e+00
cloud	shifted	8	10,000	none	1	7	78	3.062	4.8	0.00e+00
cloud	shifted	8	10,000	none	2	7	78	3.086	5.01	0.00e+00
cloud	shifted	8	10,000	none	3	7	78	3.108	4.93	0.00e+00
cloud	shifted	8	10,000	none	4	7	78	3.099	4.71	0.00e+00
cloud	shifted	8	10,000	none	5	7	78	3.080	4.81	0.00e+00
cloud	lattice_ties	8	10,000	none	1	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	2	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	3	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	4	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	5	2	78	1.170	1.1	0.00e+00
cloud	heavy_tailed	8	10,000	none	1	2	78	1.170	0.894	0.00e+00
cloud	heavy_tailed	8	10,000	none	2	2	78	1.170	0.89	0.00e+00
cloud	heavy_tailed	8	10,000	none	3	2	78	1.170	0.914	0.00e+00
cloud	heavy_tailed	8	10,000	none	4	2	78	1.170	0.894	0.00e+00
cloud	heavy_tailed	8	10,000	none	5	2	78	1.170	0.893	0.00e+00
cloud	contested	8	10,000	none	1	10	78	73.287	133	0.00e+00
cloud	contested	8	10,000	none	2	9	78	73.318	133	0.00e+00
cloud	contested	8	10,000	none	3	10	78	73.347	134	0.00e+00
cloud	contested	8	10,000	none	4	10	78	73.355	134	0.00e+00
cloud	contested	8	10,000	none	5	9	78	73.312	133	0.00e+00
cloud	shell	8	10,000	none	1	4	78	1.172	1.94	0.00e+00
cloud	shell	8	10,000	none	2	7	78	1.200	3.26	0.00e+00
cloud	shell	8	10,000	none	3	6	78	2.825	3.4	0.00e+00
cloud	shell	8	10,000	none	4	5	78	1.183	2.63	0.00e+00
cloud	shell	8	10,000	none	5	5	78	1.176	2.5	0.00e+00
caliper	gaussian	8	10,000	none	1	2	78	1.170	0.862	0.00e+00
caliper	gaussian	8	10,000	none	2	2	78	1.170	0.862	0.00e+00
caliper	gaussian	8	10,000	none	3	2	78	1.170	0.862	0.00e+00
caliper	gaussian	8	10,000	none	4	2	78	1.170	0.866	0.00e+00
caliper	gaussian	8	10,000	none	5	2	78	1.170	0.861	0.00e+00
caliper	gaussian	8	10,000	boundary	1	2	78	1.151	0.911	0.00e+00
caliper	gaussian	8	10,000	boundary	2	2	78	1.145	0.901	0.00e+00
caliper	gaussian	8	10,000	boundary	3	2	78	1.121	0.844	0.00e+00
caliper	gaussian	8	10,000	boundary	4	2	78	1.168	1.05	0.00e+00
caliper	gaussian	8	10,000	boundary	5	2	78	1.165	1	0.00e+00

Sweep	Cloud	Dim	Units	Caliper	Seed	Rounds	Seed width	Graph %	Seconds	Gap
caliper	gaussian	8	10,000	mid	1	2	78	1.166	1.04	0.00e+00
caliper	gaussian	8	10,000	mid	2	2	78	1.125	0.852	0.00e+00
caliper	gaussian	8	10,000	mid	3	2	78	1.142	0.883	0.00e+00
caliper	gaussian	8	10,000	mid	4	2	78	1.167	1.02	0.00e+00
caliper	gaussian	8	10,000	mid	5	2	78	1.163	0.991	0.00e+00
size	gaussian	8	2,000	none	1	2	66	4.951	0.05	0.00e+00
size	gaussian	8	2,000	none	2	2	66	4.951	0.05	0.00e+00
size	gaussian	8	2,000	none	3	2	66	4.951	0.051	0.00e+00
size	gaussian	8	5,000	none	1	2	72	2.160	0.248	0.00e+00
size	gaussian	8	5,000	none	2	2	72	2.160	0.25	0.00e+00
size	gaussian	8	5,000	none	3	2	72	2.160	0.25	0.00e+00
size	gaussian	8	10,000	none	1	2	78	1.170	0.866	0.00e+00
size	gaussian	8	10,000	none	2	2	78	1.170	0.861	0.00e+00
size	gaussian	8	10,000	none	3	2	78	1.170	0.863	0.00e+00
size	gaussian	8	20,000	none	1	2	84	0.630	3	0.00e+00
size	gaussian	8	20,000	none	2	2	84	0.630	2.98	0.00e+00
size	gaussian	8	20,000	none	3	2	84	0.630	2.96	0.00e+00
size	gaussian	8	50,000	none	1	2	96	0.288	15.2	0.00e+00
size	gaussian	8	50,000	none	2	3	96	0.288	21.9	0.00e+00
size	gaussian	8	50,000	none	3	3	96	0.288	21.9	3.64e-12

Every cell of the grid finished.

4 Peak memory

Peak resident set size is a property of a process, so each cell here is a fresh R process that loads what it needs, does one matching and exits, measured from outside by the platform's `time` utility. The baseline row of a size is a process that loaded the same packages and matched nothing, and it is what the increment column is measured against. `dense_matrix` builds the distance matrix and stops, which separates the representation from the solver that runs on it.

Table 10: Peak resident set size by arm and problem size. Matrix is the bytes the dense cost matrix itself needs at that shape. Package estimate is what `estimate_dense_matrix_mb()` predicts, which is four times the matrix and is what the memory dispatcher warns from.

Units	Arm	Peak RSS	Over baseline	R heap peak	Matrix	Package estimate	Seconds	Status
2,000	baseline	146 MB	-	80 MB	7 MB	28 MB	0.008	ok
2,000	baseline_alt	153 MB	-	87 MB	7 MB	28 MB	0.008	ok
2,000	dense	266 MB	120 MB	153 MB	7 MB	28 MB	0.167	ok
2,000	dense_matrix	203 MB	58 MB	137 MB	7 MB	28 MB	0.069	ok
2,000	implicit	157 MB	12 MB	102 MB	7 MB	28 MB	0.07	ok

Units	Arm	Peak RSS	Over baseline	R heap peak	Matrix	Package estimate	Seconds	Status
2,000	lazy	151 MB	6 MB	102 MB	7 MB	28 MB	0.068	ok
2,000	MatchIt	531 MB	378 MB	311 MB	7 MB	28 MB	2.35	ok
2,000	optmatch	483 MB	331 MB	318 MB	7 MB	28 MB	1.85	ok
5,000	baseline	146 MB	-	80 MB	44 MB	178 MB	0.007	ok
5,000	baseline__alt	153 MB	-	87 MB	44 MB	178 MB	0.008	ok
5,000	dense	563 MB	417 MB	409 MB	44 MB	178 MB	0.887	ok
5,000	dense__matrix	226 MB	81 MB	161 MB	44 MB	178 MB	0.339	ok
5,000	implicit	175 MB	29 MB	104 MB	44 MB	178 MB	0.315	ok
5,000	lazy	154 MB	8 MB	104 MB	44 MB	178 MB	0.365	ok
5,000	MatchIt	2436 MB	2283 MB	1474 MB	44 MB	178 MB	16.6	ok
5,000	optmatch	2059 MB	1906 MB	1320 MB	44 MB	178 MB	13.3	ok
10,000	baseline	149 MB	-	80 MB	178 MB	711 MB	0.007	ok
10,000	baseline__alt	153 MB	-	87 MB	178 MB	711 MB	0.008	ok
10,000	dense	1429 MB	1280 MB	812 MB	178 MB	711 MB	3.13	ok
10,000	dense__matrix	442 MB	294 MB	374 MB	178 MB	711 MB	1.21	ok
10,000	implicit	207 MB	58 MB	108 MB	178 MB	711 MB	1.11	ok
10,000	lazy	159 MB	10 MB	108 MB	178 MB	711 MB	1.61	ok
10,000	MatchIt	8862 MB	8709 MB	6382 MB	178 MB	711 MB	74	ok
10,000	optmatch	7344 MB	7192 MB	4545 MB	178 MB	711 MB	60.1	ok
20,000	baseline	146 MB	-	80 MB	711 MB	2844 MB	0.007	ok
20,000	baseline__alt	153 MB	-	87 MB	711 MB	2844 MB	0.007	ok
20,000	dense	6231 MB	6085 MB	3656 MB	711 MB	2844 MB	10.6	ok
20,000	dense__matrix	1589 MB	1443 MB	1145 MB	711 MB	2844 MB	4.28	ok
20,000	implicit	273 MB	127 MB	116 MB	711 MB	2844 MB	3.93	ok
20,000	lazy	166 MB	20 MB	116 MB	711 MB	2844 MB	6.42	ok
20,000	MatchIt	8793 MB	8640 MB	7121 MB	711 MB	2844 MB	113	error: (from optmatch) result would exceed 2^31-1 bytes

Units	Arm	Peak RSS	Over baseline	R heap peak	Matrix	Package estimate	Seconds	Status
20,000	optmatch	27570 MB	27417 MB	19331 MB	711 MB	2844 MB	278	ok

Table 11: The four-fold estimate against the measurement. The estimate is deliberately conservative: it exists to avoid under-warning before a large allocation, not to predict the peak.

Units	Matrix	Measured over baseline	Measured / matrix	Package estimate
2,000	7 MB	58 MB	8.13	28 MB
5,000	44 MB	81 MB	1.82	178 MB
10,000	178 MB	294 MB	1.65	711 MB
20,000	711 MB	1443 MB	2.03	2844 MB

5 Representation against solver

The implicit mode is a representation and a solver at once: its restricted master is the flow model, and naming a solver under it is an error rather than a setting. Reading its timing as a representation comparison would therefore attribute the flow solver’s behaviour to the representation. The table below is the decomposition: down a column, one solver over two representations; across a row, one representation under two solvers.

Table 12: Wall-clock seconds by representation and solver. Dense and lazy differ only in whether the matrix is materialised; dense and dense-flow differ only in the solver; implicit is the mode as a user meets it.

Units	Dense, JV	Lazy, JV	Dense, flow	Implicit
500	21 ms	8 ms	20 ms	11 ms
2,000	156 ms	53 ms	191 ms	55 ms
5,000	814 ms	348 ms	1.12 s	294 ms
10,000	3.07 s	1.59 s	4.07 s	1.1 s
20,000	not run	6.38 s	not run	3.91 s
50,000	not run	62.1 s	not run	20.2 s

6 Scaling, every run

The scaling table in the article reports the median across instances. These are the runs behind it, one row per repetition.

Table 13: Every scaling run. Instance is an independently generated problem of that size; rep is a repetition on that problem. A run that failed or timed out carries its status and no time.

Units	Package	Instance	Seed	Rep	Seconds	Status
500	couplr	1	20261015	1	0.101	ok
500	couplr	1	20261015	2	0.01	ok
500	couplr	1	20261015	3	0.011	ok

Units	Package	Instance	Seed	Rep	Seconds	Status
500	couplr	2	21261018	1	0.011	ok
500	couplr	2	21261018	2	0.011	ok
500	couplr	2	21261018	3	0.011	ok
500	couplr	3	22261021	1	0.011	ok
500	couplr	3	22261021	2	0.011	ok
500	couplr	3	22261021	3	0.01	ok
500	couplr	4	23261024	1	0.011	ok
500	couplr	4	23261024	2	0.011	ok
500	couplr	4	23261024	3	0.011	ok
500	couplr	5	24261027	1	0.011	ok
500	couplr	5	24261027	2	0.011	ok
500	couplr	5	24261027	3	0.009	ok
500	MatchIt	1	20261015	1	0.11	ok
500	MatchIt	1	20261015	2	0.112	ok
500	MatchIt	1	20261015	3	0.113	ok
500	MatchIt	2	21261018	1	0.115	ok
500	MatchIt	2	21261018	2	0.113	ok
500	MatchIt	2	21261018	3	0.151	ok
500	MatchIt	3	22261021	1	0.116	ok
500	MatchIt	3	22261021	2	0.117	ok
500	MatchIt	3	22261021	3	0.117	ok
500	MatchIt	4	23261024	1	0.121	ok
500	MatchIt	4	23261024	2	0.124	ok
500	MatchIt	4	23261024	3	0.122	ok
500	MatchIt	5	24261027	1	0.115	ok
500	MatchIt	5	24261027	2	0.117	ok
500	MatchIt	5	24261027	3	0.117	ok
500	optmatch	1	20261015	1	0.097	ok
500	optmatch	1	20261015	2	0.13	ok
500	optmatch	1	20261015	3	0.094	ok
500	optmatch	2	21261018	1	0.097	ok
500	optmatch	2	21261018	2	0.098	ok
500	optmatch	2	21261018	3	0.13	ok
500	optmatch	3	22261021	1	0.098	ok
500	optmatch	3	22261021	2	0.099	ok
500	optmatch	3	22261021	3	0.098	ok
500	optmatch	4	23261024	1	0.107	ok
500	optmatch	4	23261024	2	0.139	ok
500	optmatch	4	23261024	3	0.104	ok
500	optmatch	5	24261027	1	0.099	ok
500	optmatch	5	24261027	2	0.099	ok
500	optmatch	5	24261027	3	0.136	ok
2,000	couplr	1	20262515	1	0.107	ok
2,000	couplr	1	20262515	2	0.139	ok
2,000	couplr	1	20262515	3	0.142	ok
2,000	couplr	2	21262518	1	0.106	ok
2,000	couplr	2	21262518	2	0.105	ok
2,000	couplr	2	21262518	3	0.109	ok
2,000	couplr	3	22262521	1	0.112	ok
2,000	couplr	3	22262521	2	0.146	ok
2,000	couplr	3	22262521	3	0.114	ok
2,000	couplr	4	23262524	1	0.112	ok

Units	Package	Instance	Seed	Rep	Seconds	Status
2,000	couplr	4	23262524	2	0.11	ok
2,000	couplr	4	23262524	3	0.11	ok
2,000	couplr	5	24262527	1	0.109	ok
2,000	couplr	5	24262527	2	0.112	ok
2,000	couplr	5	24262527	3	0.11	ok
2,000	MatchIt	1	20262515	1	2.076	ok
2,000	MatchIt	1	20262515	2	2.105	ok
2,000	MatchIt	1	20262515	3	2.103	ok
2,000	MatchIt	2	21262518	1	2	ok
2,000	MatchIt	2	21262518	2	2.029	ok
2,000	MatchIt	2	21262518	3	1.999	ok
2,000	MatchIt	3	22262521	1	2.152	ok
2,000	MatchIt	3	22262521	2	2.175	ok
2,000	MatchIt	3	22262521	3	2.231	ok
2,000	MatchIt	4	23262524	1	2.119	ok
2,000	MatchIt	4	23262524	2	2.107	ok
2,000	MatchIt	4	23262524	3	2.094	ok
2,000	MatchIt	5	24262527	1	2.076	ok
2,000	MatchIt	5	24262527	2	1.954	ok
2,000	MatchIt	5	24262527	3	2.035	ok
2,000	optmatch	1	20262515	1	1.689	ok
2,000	optmatch	1	20262515	2	1.716	ok
2,000	optmatch	1	20262515	3	1.673	ok
2,000	optmatch	2	21262518	1	1.589	ok
2,000	optmatch	2	21262518	2	1.583	ok
2,000	optmatch	2	21262518	3	1.603	ok
2,000	optmatch	3	22262521	1	1.77	ok
2,000	optmatch	3	22262521	2	1.733	ok
2,000	optmatch	3	22262521	3	1.736	ok
2,000	optmatch	4	23262524	1	1.698	ok
2,000	optmatch	4	23262524	2	1.684	ok
2,000	optmatch	4	23262524	3	1.707	ok
2,000	optmatch	5	24262527	1	1.596	ok
2,000	optmatch	5	24262527	2	1.654	ok
2,000	optmatch	5	24262527	3	1.606	ok
5,000	couplr	1	20265515	1	0.722	ok
5,000	couplr	1	20265515	2	0.71	ok
5,000	couplr	2	21265518	1	0.71	ok
5,000	couplr	2	21265518	2	0.681	ok
5,000	couplr	3	22265521	1	0.771	ok
5,000	couplr	3	22265521	2	0.689	ok
5,000	couplr	4	23265524	1	0.665	ok
5,000	couplr	4	23265524	2	0.649	ok
5,000	couplr	5	24265527	1	0.74	ok
5,000	couplr	5	24265527	2	0.646	ok
5,000	MatchIt	1	20265515	1	16.02	ok
5,000	MatchIt	1	20265515	2	15.39	ok
5,000	MatchIt	2	21265518	1	15.04	ok
5,000	MatchIt	2	21265518	2	15.06	ok
5,000	MatchIt	3	22265521	1	15.33	ok
5,000	MatchIt	3	22265521	2	15.19	ok
5,000	MatchIt	4	23265524	1	15.01	ok

Units	Package	Instance	Seed	Rep	Seconds	Status
5,000	MatchIt	4	23265524	2	15.07	ok
5,000	MatchIt	5	24265527	1	14.93	ok
5,000	MatchIt	5	24265527	2	14.76	ok
5,000	optmatch	1	20265515	1	13.09	ok
5,000	optmatch	1	20265515	2	12.68	ok
5,000	optmatch	2	21265518	1	12.27	ok
5,000	optmatch	2	21265518	2	12.2	ok
5,000	optmatch	3	22265521	1	12.19	ok
5,000	optmatch	3	22265521	2	12.27	ok
5,000	optmatch	4	23265524	1	12.61	ok
5,000	optmatch	4	23265524	2	12.18	ok
5,000	optmatch	5	24265527	1	11.84	ok
5,000	optmatch	5	24265527	2	11.82	ok
10,000	couplr	1	20270515	1	2.934	ok
10,000	couplr	1	20270515	2	2.858	ok
10,000	couplr	2	21270518	1	2.932	ok
10,000	couplr	2	21270518	2	2.827	ok
10,000	couplr	3	22270521	1	3.127	ok
10,000	couplr	3	22270521	2	3.041	ok
10,000	MatchIt	1	20270515	1	72.36	ok
10,000	MatchIt	1	20270515	2	73.67	ok
10,000	MatchIt	2	21270518	1	72.74	ok
10,000	MatchIt	2	21270518	2	72.92	ok
10,000	MatchIt	3	22270521	1	70.6	ok
10,000	MatchIt	3	22270521	2	72.44	ok
10,000	optmatch	1	20270515	1	59.41	ok
10,000	optmatch	1	20270515	2	58.93	ok
10,000	optmatch	2	21270518	1	59.27	ok
10,000	optmatch	2	21270518	2	59.54	ok
10,000	optmatch	3	22270521	1	58.28	ok
10,000	optmatch	3	22270521	2	57.67	ok
20,000	couplr	1	20280515	1	10.49	ok
20,000	couplr	2	21280518	1	19.25	ok
20,000	MatchIt	1	20280515	1	-	error: (from optmatch) result would exceed 2 ³¹ -1 bytes
20,000	MatchIt	2	21280518	1	-	error: (from optmatch) result would exceed 2 ³¹ -1 bytes
20,000	optmatch	1	20280515	1	278.8	ok
20,000	optmatch	2	21280518	1	291.1	ok
50,000	couplr	1	20310515	1	109.3	ok
50,000	MatchIt	1	20310515	1	-	timeout
50,000	optmatch	1	20310515	1	-	timeout

7 Software design

The article describes the three layers the package is organised in, the dispatch rules, the memory modes and how the package is verified. This section carries the object model, the C++ organisation and the dependency footprint.

7.1 Object model

The object model is S3 throughout. Matching results carry the class vector `c("matching_result", "couplr_result")`, with the design-specific classes (`full_matching_result`, `cem_result`, `subclass_result`) sharing the `couplr_result` parent, which is what lets `balance_diagnostics()`, `join_matched()`, `match_data()` and the `bal.tab()` methods dispatch on it once rather than being rewritten per design. Results are plain lists of tibbles, so a user who wants something the package does not provide can reach in and take it. `lap_solve()` returns a tibble subclassed as `lap_solve_result` with `source`, `target` and `cost` columns, so an optimum is usable in a plot or a join without an extraction step, while `get_total_cost()` and `get_method_used()` recover the metadata carried in attributes.

7.2 C++ organisation

All nineteen solvers are written from scratch in C++ and exposed through Rcpp, with `src/core` holding the cost-source types and shared utilities with no Rcpp dependency, `src/solvers` one algorithm per file with a thin `_rcpp.cpp` wrapper each, `src/flow` the flow model, its two solvers and the edge-generation loop, and `src/interface` the conversion between R objects and the internal types. The separation means the solver bodies are ordinary C++ that can be compiled and tested outside R, which is what the C++ test suite does.

Solvers are templated on their cost source rather than written against a concrete matrix. A cost source provides `at(i, j)`, `allowed(i, j)` and `empty()`, and two types satisfy that interface: `CostMatrix`, which owns a dense flat array plus a mask, and `LazyCostMatrix`, which owns the two feature matrices and computes C_{ij} on demand from the requested metric, applying calipers and the distance ceiling in `allowed()`. A handful of hot loops index the dense mask array directly for speed, which the lazy type cannot support; these are guarded by a `supports_raw_mask` trait and an `if constexpr` branch rather than by duplicating the solver. One solver body therefore serves both cost representations, and the pricing oracle of the edge-generation loop needs `at(i, j)` and `allowed(i, j)` and nothing else, so it consumes the same concept and adds no third representation of the costs.

`LazyCostMatrix` also bakes in the transformations that the dense path applies in a separate preparation pass, namely mapping forbidden entries to a large sentinel and negating costs when maximising. Doing this at construction is what allows the untouched solver body to work on either type.

7.3 Dependency footprint

`couplr` declares Rcpp, tibble, dplyr, rlang, purrr and `methods` in `Imports`, and Rcpp alone in `LinkingTo`. The recursive closure of hard dependencies is 17 non-base packages, so a default installation brings 18 packages including `couplr` itself. The comparable figures on the same day were 18 for `optmatch`, 8 for `MatchIt` and 7 for `designmatch`.

Two choices sit behind that number. First, we use no external solver: all nineteen algorithms are package source, so there is no dependency on an LP or MIP library and no system solver to install. That choice also keeps `LinkingTo` to Rcpp alone, which matters because every entry there forces a compile of that package before `couplr` can be built from source; `couplr` declares only what its headers actually include. Second, everything that serves a single feature lives in `Suggests` and is loaded at call time. The animation and image-morphing functions, the plotting helpers, the interoperability layer for `MatchIt`, `optmatch` and `cobalt`, and the parallel back end are all optional in this sense, so a user who only wants to match pays for none of them.